

Segmented Arbitrage

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ABSTRACT

We use arbitrage activity in equity, fixed income, and foreign exchange markets to characterize the frictions and constraints facing intermediaries. The average pairwise correlation between the 32 arbitrage spreads that we study is 22%. These low correlations are inconsistent with canonical intermediary asset pricing models. We show that at least two types of segmentation drive arbitrage dynamics. First, funding is segmented—certain trades rely on specific funding sources, making their arbitrage spreads sensitive to localized funding shocks. Second, balance sheets are segmented—intermediaries specialize in certain trades, so arbitrage spreads are sensitive to idiosyncratic balance-sheet shocks.

THERE IS GROWING RECOGNITION THAT financial intermediaries play a key role in determining asset prices. Much of the research on intermediaries treats them as a monolith, assuming that all financial institutions face the same—typically limited—set of constraints, fund homogeneously from the household sector, and perfectly share risk with each other. This view of intermediaries has several implications. In particular, it suggests that all risk premia should strongly comove with aggregate intermediary balance-sheet strength, and conversely that all risk premia should be equally informative about the health of the intermediary sector. In addition, if intermediaries are strongly integrated, fire sales in any market have economy-wide effects on credit creation because intermediaries will reduce lending and instead provide that market with liquidity.

In this paper, we argue that the assumption of a representative intermediary, while helpful for many applications, understates the importance of frictions within the intermediary sector and their implications for prices. We

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provide empirical evidence that segmentation within the intermediary sector has a first-order impact on asset prices. We focus our analysis on arbitrage spreads—riskless returns in excess of riskless rates—that arise from violations of the law of one price in equity, fixed income, and foreign exchange markets. We take this approach for two reasons. First, arbitrage is intermediated by financial institutions such as broker-dealers and hedge funds and cannot be easily performed by households (Haddad and Muir (2021)). Second, arbitrage spreads are accurate measures of expected returns, the key objects in any asset pricing theory, and thus arbitrages offer a high-power setting for understanding the frictions faced by intermediaries. In contrast, studies analyzing risky assets must rely on average realized returns, which are noisy proxies for expected returns (Merton (1980)).

To fix ideas, we begin with a stylized model in which intermediaries determine arbitrage spreads. In the model, a continuum of intermediaries participates in a set of fundamentally riskless arbitrage trades. Intermediaries potentially face two types of frictions that break the Modigliani and Miller (1958) theorem. First, they may face balance-sheet constraints like regulatory capital requirements, which are costly to satisfy due to external financing frictions. Second, intermediaries may face frictions that prevent them from raising financing to fund riskless assets at the riskless rate. Intermediaries may fund from different sources with different costs, and certain trades may require them to fund from a specific source. We model these frictions in reduced form to focus on their implications for arbitrage spreads.

In the model and throughout the paper, we distinguish between three assumptions typically embedded in theoretical and applied work using a representative intermediary. First, balance-sheet integration means that the marginal balance-sheet cost of holding a given riskless asset is equalized across intermediaries. Second, funding integration means intermediaries can fund all riskless assets from the same source. Third, the set of constraints that intermediaries face is limited. These assumptions result in one- or two-factor structures for arbitrage spreads.¹ For instance, if the representative intermediary faces a single constraint (e.g., a leverage constraint) and funding is frictionless, then all arbitrage spreads are determined by the shadow cost of the constraint. In this case, spreads are perfectly correlated and follow a single-factor structure.

We then use the model to illustrate how segmentation can reduce correlations between arbitrage spreads. Funding segmentation—violations of funding integration—can reduce correlations between trades that use different funding sources. Similarly, balance-sheet segmentation—violations of balance-sheet integration—implies that trades performed by the same arbitrageurs will be more correlated with each other than trades performed by different arbitrageurs.

¹ See, for example, He, Kelly, and Manela (2017), He and Krishnamurthy (2013), Adrian, Etula, and Muir (2014), Ivashina, Scharfstein, and Stein (2015), Gromb and Vayanos (2018), and Andersen, Duffie, and Song (2019).

We next turn to the data, focusing on the decade following the 2007 to 2009 Global Financial Crisis (GFC). We study 32 arbitrage trades that fall into seven broad strategies: (i) equity spot-futures arbitrage, (ii) equity options arbitrage, which enforces put-call parity, (iii) currency spot-futures arbitrage, which enforces covered interest parity (CIP), (iv) credit default swap (CDS)-bond arbitrage, (v) Treasury spot-futures arbitrage, (vi) Treasury-interest rate swaps arbitrage, and (vii) Treasury-inflation swaps arbitrage. For each arbitrage trade, we define the spread as the difference between the riskless rate implied by no-arbitrage conditions (e.g., spot-futures parity) and a relevant benchmark rate.

Our first result is that the daily correlation of spreads is low on average. The average pairwise correlation is 0.22, and the 75th percentile of pairwise correlations is 0.42. While these low correlations could be driven by measurement error, this measurement error would have to be large to explain our results since observed correlations are far from one. Furthermore, we observe a similar factor structure if we smooth the data. For instance, after taking monthly moving averages, 10 principal components (PCs) are required to explain 90% of the variation in arbitrage spreads. Correlations are also low among the subsample of arbitrage trades with short tenors (three- to six-month horizons), suggesting that convergence or noise trader risk (DeLong et al. (1990)) do not drive the high-dimensional factor structure. The data are therefore far from the one- or two-factor structure predicted by models in which balance-sheet and funding integration hold in an intermediary sector facing few constraints.

Departures from the integrated intermediary benchmark imply that the correlations of arbitrage spreads are determined by both arbitrageur supply and end-user demand. One might therefore wonder whether the low correlations we document arise primarily from the supply side or the demand side. We use the sign-restricted structural vector autoregression (SVAR) methodology of Uhlig (2005) to uncover separate supply and demand shocks for the subset of trades for which we have quantity data. Correlations of supply shocks across trades are low, consistent with the idea that intermediaries performing these arbitrages face significant frictions.

We next show that funding segmentation is one reason that correlations between arbitrage spreads are low. Our analysis starts from the observation that equity spot-futures, equity options, and CIP arbitrage face relatively higher margin requirements than other strategies. Because these high-margin strategies require more unsecured funding, we refer to them as “unsecured” arbitrages, while we refer to the remaining ones as “secured” arbitrages.² Unsecured arbitrages are more correlated with each other than they are with secured arbitrages. We provide evidence that this higher correlation reflects the higher exposure of unsecured arbitrages to conditions in unsecured funding markets, which we proxy for using the Treasury-Eurodollar (TED)

² Secured arbitrages include Treasury spot-futures, Treasury-swap, Treasury Inflation-Protected Securities (TIPS)-Treasury, and CDS-bond.

spread. We find that unsecured arbitrage spreads are nearly seven times more sensitive to movements in the TED spread than are secured arbitrage spreads.

While the higher loading of unsecured arbitrage spreads on the TED spread is consistent with funding segmentation, it could also be driven by balance-sheet segmentation. For example, if broker-dealers specialize in unsecured arbitrages, then a deterioration of their balance sheets could cause both the TED spread and unsecured arbitrage spreads to rise. To isolate the role of funding segmentation, it is therefore useful to trace out how shocks to the supply of unsecured funding differentially impact unsecured versus secured arbitrages. Following Anderson, Du, and Schlusche (2019), we conduct an event study around the 2016 money market mutual fund (MMF) reform, which resulted in a sharp contraction in unsecured lending by MMFs. During the reform, the TED spread and unsecured arbitrage spreads rise, while secured arbitrage spreads do not, demonstrating that segmentation in funding markets is an important driver of arbitrage spreads.

We also provide evidence that funding markets are more segmented than the simple divide between secured and unsecured trades because funding providers specialize (Chernenko and Sunderam (2014), Li (2021)). Thus, shocks to individual funding sources move specific arbitrage spreads without moving others. We illustrate this idea by studying supply shocks to Fidelity MMFs, which Hu, Pan, and Wang (2021) show are especially active in funding holders of equity securities. These shocks move equity spot-futures arbitrage spreads but not others.

We next show that balance-sheet segmentation also contributes to the low overall correlation of arbitrage spreads. In other words, it is not the case that intermediary balance sheets are integrated and a representative intermediary facing segmented funding is marginal in all strategies. We first provide event-study evidence that the balance-sheet constraints of certain intermediaries affect some trades more than others. We study the “London Whale” episode, in which JPMorgan lost over \$6 billion through its credit derivatives hedging program in 2012. This event is useful for our purposes because it did not materially affect the firm’s funding rates but did result in a tightening of the firm’s risk limits (U.S. Senate (2013)). We show that the episode led equity spot-futures arbitrage spreads to rise relative to others.

Balance-sheet segmentation is also evidenced by the fact that secured arbitrage spreads tend to rise following losses at hedge funds that specialize in fixed-income arbitrage, yet unsecured arbitrage spreads do not. These joint patterns suggest that hedge fund balance sheets are particularly important for secured arbitrages. Moreover, specific hedge funds appear to matter for specific trades. For example, the hedge funds with balance sheets important for CDS-bond arbitrage are not the ones that are important for TIPS-Treasury arbitrage. Overall, our evidence suggests that arbitrage activity is segmented due to fragmented funding sources (e.g., unsecured vs. secured) and specialization across financial institutions (e.g., dealers vs. hedge funds).

Our paper belongs to the rapidly expanding literature on financial intermediaries and their role in capital markets. Prior theoretical work in this

area, including Shleifer and Vishny (1997), Gromb and Vayanos (2002), Brunnermeier and Pedersen (2009), Gârleanu and Pedersen (2011), Adrian and Boyarchenko (2012), He and Krishnamurthy (2013), and Brunnermeier and Sannikov (2014), generally assumes a representative intermediary and studies how different constraints on its activity impact equilibrium asset prices or arbitrage spreads. Our results suggest that these theories most naturally describe market segments, rather than provide a uniform account of dynamics across all capital markets.³

The empirical literature can be divided into three categories. One strand studies law-of-one-price violations in specific markets, including equity (van Binsbergen, Diamond, and Grotteria (2019), Hazelkorn, Moskowitz, and Vasudevan (2023)), foreign exchange (Du, Tepper, and Verdelhan (2018)), Treasury (Fleckenstein, Longstaff, and Lustig (2014); Jerermann (2020); Barth and Kahn (2021)), and corporate bond markets (Bai and Collin-Dufresne (2019)). Our paper departs from this research by simultaneously analyzing law-of-one-price violations across many different markets, which enables us to characterize the frictions and constraints faced by different intermediaries.⁴ A second strand, including Pasquariello (2014), Adrian, Etula, and Muir (2014), He, Kelly, and Manela (2017), and Du, Hébert, and Huber (2019), aims to empirically link sector-level measures of intermediary constraints to risky asset prices.⁵ Our results suggest that accounting for which intermediaries are active in a market and how they fund themselves is likely to improve the performance of these kinds of intermediary-based asset pricing models. This conclusion accords with Siriwardane (2019), who shows how specialization by financial intermediaries affects the pricing of credit derivatives.

A third strand of the empirical literature, including Boyarchenko et al. (2016, 2020) and Liu (2020), studies large panels of arbitrage spreads as we do. These papers emphasize moments of the data that are consistent with the integrated intermediary model. For instance, Boyarchenko et al. (2016, 2020) assume that globally systemically important banks are all active in a wide range of arbitrages and study how spreads respond to the introduction of the supplementary leverage ratio requirement under this assumption. Similarly, Liu (2020) highlights the common variation in spreads by studying their first PC and correlating it with the TED spread and hedge fund returns. In contrast, our analysis starts from moments that are inconsistent with the

³ A recent theoretical literature has emphasized the importance of intermediary heterogeneity for macroeconomic outcomes and optimal macroprudential policy (Begenau and Landvoigt (2021); Jamilov (2021)).

⁴ Prior work also documents segmentation in short-term money markets (Bech and Klee (2011); Duffie and Krishnamurthy (2016)). Our paper shows how that segmentation ultimately impacts risky asset prices.

⁵ Adrian, Etula, and Muir (2014) and He, Kelly, and Manela (2017) fail to reject the null of integration based on a test of whether the prices of risk for intermediary factors differ across markets. However, their tests use realized average returns to proxy for ex ante risk premia, which lowers their power. Accordingly, Bryzgalova (2015) finds that the quarterly intermediary capital factor is weak in the sense that it has a small covariance with asset returns.

integrated intermediary model, namely the pairwise correlation matrix of arbitrage spreads. This difference in empirical focus leads us to conceptually different economic conclusions from the existing literature. We start from moments that suggest segmentation and then characterize the nature of that segmentation, emphasizing frictions in funding markets and the segmentation of balance sheets (i.e., arbitrageur equity capital) as two distinct drivers of low correlations between arbitrage spreads.

The remainder of the paper proceeds as follows. Section I introduces a stylized model that is used to organize our empirical analysis. Section II presents our main fact—the low correlation among arbitrage spreads—and demonstrates that it is unlikely to be driven by noise trader risk or arbitrageur inactivity. An SVAR further shows that the factor structure of arbitrage spreads is in large part attributable to weakly correlated supply shocks. Sections III and IV trace this high-dimensional factor structure to funding and balance-sheet segmentation. Section V concludes.

I. Motivating Model

To fix ideas, we begin with a stylized model in which intermediaries face multiple frictions and determine arbitrage spreads. The model highlights how balance-sheet constraints, balance-sheet segmentation, and funding segmentation all impact arbitrage spreads. The key point is that the three assumptions typical of the intermediary asset pricing literature—(i) a small number of constraints, (ii) balance-sheet integration, and (iii) funding integration—result in highly correlated arbitrage spreads. Violating any of the three assumptions can result in the high-dimensional factor structure for arbitrage spreads that we document below. Balance-sheet and funding segmentation further predict that some spreads move with proxies for balance-sheet and funding costs, but others do not.

A. Setup

Formally, suppose there are N arbitrage trades that are riskless. Normalize the riskless rate to zero and let $s_{n,t}$ denote the arbitrage spread on trade n at time t . For simplicity, we assume that arbitrageurs are always net long, so that all spreads in the model are positive. In the empirical analysis below, we work with the absolute value of spreads since arbitrageurs can be net long or net short each trade.

A unit measure of competitive and atomistic arbitrageurs (i.e., intermediaries) engages in these trades, supplying $q_{n,t}$ of trade n .⁶ Arbitrageurs face two main frictions, both of which are modeled in reduced form. First, there are K balance-sheet requirements of the form $\sum_n q_{n,t} v_{n,k} = V_{k,t}$. These requirements

⁶ As discussed in Wallen (2019), market power among intermediaries may be important in certain markets. The results here would be qualitatively unchanged in oligopolistic market structures if the elasticity of demand from outside investors is constant over time.

capture equity capital and liquidity constraints, which may be set by regulators or by arbitrageurs themselves for internal risk-management purposes. We assume that the contribution of trade n to constraint k , $v_{n,k}$, is fixed over time. Arbitrageurs can adjust their balance sheets to meet requirement k at total cost $\frac{1}{2}c_{k,t}V_{k,t}^2$, which capture the costs of external finance or other adjustment costs. For instance, a high value of $c_{k,t}$ could indicate that the signaling problem associated with equity issuance is more severe. The existence of balance-sheet requirements does not imply balance-sheet segmentation. Even with multiple balance-sheet requirements, all arbitrageurs can face the same marginal balance-sheet cost for a given trade, which means that we can model a single, representative intermediary for all trades. We introduce balance-sheet segmentation below.

Second, there are funding frictions. There are L funding sources with associated cost $f_{1,t}, \dots, f_{L,t}$ (in excess of the riskless rate of zero) per unit borrowed. One dollar of trade n can be financed with $w_{n,l}$ dollars from funding source $l \in L$. This assumption captures violations of the Modigliani and Miller (1958) theorem in funding markets. Despite the fact that all N trades are riskless, arbitrageurs may not be able to fund the basket of securities and derivatives that underlie each trade at the riskless rate (Duarte, Longstaff, and Yu (2007)). For instance, Treasury and equity repo rates differ even when both sources of funding are used in a fundamentally riskless futures-basis arbitrage. The assumption that $w_{n,l}$ does not vary over time corresponds to the empirical notion that rates on funding fluctuate more than haircuts (Copeland, Martin, and Walker (2010)). If $L = 0$, then funding is frictionless. If $L = 1$, then funding is frictional but integrated. Finally, if $L > 1$ (and $w_{n,l}$ varies across trades and financing sources), then funding is segmented.

The arbitrageur's problem is

$$\max \sum_{n=1}^N \left(q_{n,t} \left(s_{n,t} - \sum_l w_{n,l} f_{l,t} \right) \right) - \frac{1}{2} \sum_{k=1}^K c_{k,t} V_{k,t}^2. \quad (1)$$

Since arbitrageurs are atomistic, they take $s_{n,t}$ as given. To close the model, we assume that outside demand for trade n is inelastic and given by $a_{n,t} > 0$. Market clearing then requires that $q_{n,t} = a_{n,t}$.⁷

B. Canonical Intermediary Asset Pricing Models

Though it is stylized, the model allows us to nest common assumptions in the intermediary asset pricing literature. We discuss two typical structures here, both of which feature balance-sheet and funding integration.

Balance Sheet and Funding Integration with a Single Balance-Sheet Constraint. Many models of intermediaries consider a single balance-sheet constraint and frictionless funding (e.g., He and Krishnamurthy (2013)). This case

⁷ We make the assumption that outside demand is completely inelastic for simplicity. Our key results are qualitatively the same if outside demand is elastic (e.g., given by $a_{n,t} - b_n s_{n,t}$).

can be captured by setting $f_{l,t} = 0$ for all l , $c_{1,t} \neq 0$, and $c_{k,t} = 0$ for all $k > 1$. The solution to equation (1) is then given by

$$s_{n,t} = v_{n,1}c_{1,t}V_{1,t} = v_{n,1}c_{1,t}\left(\sum_n a_{n,t}v_{n,1}\right). \quad (2)$$

From this expression, it is clear that spreads will be perfectly correlated. There is a single factor—the marginal cost of the balance-sheet constraint, $c_{1,t}V_{1,t}$ —that moves all trades proportionally. Trades that face a higher balance-sheet requirement $v_{n,1}$ load more heavily on this factor, but all spreads move linearly with the marginal cost of the constraint. This one-factor structure in spreads holds despite the fact that there are a large number of primitive shocks in the model. In particular, the balance-sheet shocks $c_{1,t}$ and the outside demand shocks $a_{n,t}$ for each trade n fluctuate independently, yet a one-factor structure still obtains. The intuition is that these independent shocks all move the marginal cost of the balance sheet, which is ultimately all that matters for spreads.

Balance-Sheet and Funding Integration with a Single Funding Factor. Another simple structure featuring both balance-sheet and funding integration involves no constraints and a single frictional funding factor: $c_{k,t} = 0$, $v_{n,k} = 0$, $f_{n,1} > 0$, and $f_{n,l} = 0$ for $l > 1$. In this case, spreads are simply driven by the funding factor, $s_{n,t} = w_{n,1}f_{1,t}$, and we again have perfect correlations across spreads. Spreads may load differentially on the funding factor, but they all move linearly with it.⁸

C. Integration with Many Constraints

While much of the intermediary asset pricing literature features perfectly correlated arbitrage spreads, balance-sheet and funding integration need not imply them. In particular, balance-sheet and funding integration admit a single frictional funding source ($L = 1$) and arbitrarily many balance-sheet constraints ($K > 0$). In this case, all riskless arbitrages are funded from the same source and marginal balance-sheet costs are equated across arbitrageurs for each trade n . Spreads are given by

$$s_{n,t} = w_{n,1}f_{1,t} + \sum_{k=1}^K v_{n,k}c_{k,t}V_{k,t} \quad (3)$$

and feature a $K + 1$ factor structure. Thus, a high-dimensional factor structure for arbitrage spreads rules out balance-sheet and funding integration with a small number of constraints.

⁸ Andersen, Duffie, and Song (2019) has this reduced form, although formally they obtain the result by microfounding the costs of external equity with a debt overhang problem. With this microfoundation, the marginal cost of external equity funding for a riskless asset, $w_{n,equity}f_{equity,t}$, is equal to the arbitrageur's credit spread.

D. Segmentation

We next consider the impact of segmentation on spreads. We consider two types of segmentation: funding segmentation and balance-sheet segmentation.

Segmented Funding. By funding segmentation, we mean that certain trades can use certain funding sources while other trades cannot. For instance, Treasury repo financing can be used for Treasury spot-futures arbitrage but cannot be used for equity spot-futures arbitrage. To see the implications of this kind of segmentation, suppose that trades $n = 1, \dots, N_1 < N$ can be funded only using source $l = 1$ with corresponding cost $f_{1,t}$, while trades $n = N_1 + 1, \dots, N$ can be funded only using source $l = 2$ with corresponding cost $f_{2,t}$. If there are no further frictions, we have

$$s_{n,t} = \begin{cases} w_{n,1}f_{1,t} & \text{if } n \leq N_1 \\ w_{n,2}f_{2,t} & \text{if } N_1 < n. \end{cases} \quad (4)$$

In this case, spreads have a two-factor structure. All trades that can be funded using source 1 are perfectly correlated, as are all trades that can be funded using source 2, but the correlation between the two groups is the correlation between $f_{1,t}$ and $f_{2,t}$:

$$\rho(s_{n_1,t}, s_{n_2,t}) = \begin{cases} 1 & \text{if } n_1, n_2 \leq N_1 \text{ or } N_1 < n_1, n_2 \\ \rho(f_{1,t}, f_{2,t}) & \text{if } n_1 \leq N_1, n_2 > N_1. \end{cases} \quad (5)$$

Extending the argument to more than two funding sources, segmented funding can create a high-dimensional factor structure for arbitrage spreads. In this case, the outside demand shocks $a_{n,t}$ still do not impact the correlation of spreads because balance sheets are integrated and funding is elastically supplied.

Segmented Balance Sheets. Finally, we consider balance-sheet segmentation with frictionless funding. We use balance-sheet segmentation to describe environments in which certain trades are done by one set of intermediaries and are therefore subject to their balance-sheet constraints, while other trades are done by another set of intermediaries and are subject to their balance-sheet constraints. One could microfound this segmentation with a small amount of specialization in different trades. For instance, suppose there are small marginal costs $\varepsilon_{n,i}$ associated with arbitrageur i doing trade n and there are two types of arbitrageurs. Arbitrageurs $i \in I$ have a marginal cost advantage $\varepsilon_{n,i} < \varepsilon_{n,j}$ for trades $n = 1, \dots, N_1$ over all other arbitrageurs $j \in \sim I$. Conversely, arbitrageurs $j \in \sim I$ have a marginal cost advantage $\varepsilon_{n,j} < \varepsilon_{n,i}$ for trades $n = N_1 + 1, \dots, N$. In other words, one group of arbitrageurs has a cost advantage in one set of trades, while the other has a cost advantage in a different set of trades. There is a representative arbitrageur for each group.

Finally, suppose there is a single balance-sheet constraint ($f_{l,t} = 0$ for all l , $c_{1,t} \neq 0$, and $c_{k,t} = 0$). Further allow arbitrageurs $i \in I$ to face a different frictional cost of meeting the single balance-sheet constraint, $\frac{1}{2}c_{1,t}^I (V_{1,t}^I)^2$ than arbitrageurs $j \notin I$, $\frac{1}{2}c_{1,t}^{\sim I} (V_{1,t}^{\sim I})^2$. If the outside demand for each group of

trades is similar, then by market clearing arbitrageurs $i \in I$ will absorb all demand for trades $n = 1, \dots, N_1$ and arbitrageurs $j \in \sim I$ will absorb all demand for trades $n = N_1 + 1, \dots, N$.⁹ This implies that $V_{1,t}^I = \sum_{n=1}^{N_1} a_{n,t} v_{n,1}$ and $V_{1,t}^{\sim I} = \sum_{n=N_1+1}^N a_{n,t} v_{n,1}$, where $a_{n,t}$ is the outside demand for trade n .

In this case, spreads are given by

$$s_{n,t} = \begin{cases} \varepsilon_{n,i} + v_{n,1} c_{1,t}^I V_{1,t}^I & \text{if } n \leq N_1 \\ \varepsilon_{n,j} + v_{n,1} c_{1,t}^{\sim I} V_{1,t}^{\sim I} & \text{if } N_1 < n. \end{cases} \quad (6)$$

In other words, spreads have a two-factor structure. Intuitively, spreads for the first group of trades ($n = 1, \dots, N_1$) reflect the shadow cost of the balance-sheet constraint for arbitrageurs in group I . For the second group of trades ($n = N_1 + 1, \dots, N$), spreads will reflect the shadow cost of the balance-sheet constraint for arbitrageurs outside group I .

To get closed-form expressions for the correlation structure in this case, assume that balance-sheet costs $c_{1,t}^I$ and $c_{1,t}^{\sim I}$, as well as all outside demand shocks are jointly normally distributed with mean zero. For simplicity, also assume that there is a single trade in each group: group I intermediates trade 1 and group $\sim I$ intermediates trade 2. Then we can use the results in Bohrnstedt and Goldberger (1969) to compute the correlations of spreads,

$$\rho(s_{1,t}, s_{2,t}) = \rho(c_{1,t}^I, c_{1,t}^{\sim I}) \rho(a_{1,t}, a_{2,t}) + \rho(c_{1,t}^I, a_{2,t}) \rho(c_{1,t}^{\sim I}, a_{1,t}). \quad (7)$$

This equation says the correlation between two spreads intermediated by different groups of arbitrageurs depends on (i) the correlation of balance-sheet shocks $\rho(c_{1,t}^I, c_{1,t}^{\sim I})$, (ii) the correlation of demand shocks $\rho(a_{1,t}, a_{2,t})$, and (iii) cross terms, that is, the correlation of balance-sheet shocks for one group of arbitrageurs with the demand shocks facing another group of arbitrageurs, $\rho(c_{1,t}^I, a_{2,t})$ and $\rho(c_{1,t}^{\sim I}, a_{1,t})$. This intuition extends to the case in which each group of arbitrageurs intermediates many trades. In that case, the correlation of spreads intermediated by the same group of arbitrageurs is 1, while the correlation of spreads intermediated by different groups is determined by correlations of supply, demand, and cross terms.

This setup can also be used to explore the implications of slow-moving capital for the correlation of spreads. Suppose without loss of generality that the marginal cost of the balance sheet is higher for arbitrageurs in group I : $c_{1,t}^I V_{1,t}^I > c_{1,t}^{\sim I} V_{1,t}^{\sim I}$. Further suppose that between times t and $t + \Delta t$, flows equilibrate the marginal costs of capital between the two groups. For instance, suppose that the marginal cost advantages $\varepsilon_{n,i}$ between the two groups vanish between t and $t + \Delta t$. Then arbitrageurs in group I could sell some of their positions to other arbitrageurs, and we have $c_{1,t+\Delta t}^I V_{1,t+\Delta t}^I = c_{1,t+\Delta t}^{\sim I} V_{1,t+\Delta t}^{\sim I}$. In other words, with such capital flows, we get back to the integrated intermediary model and spreads are again perfectly correlated. Extending the argument, if capital flows are such that differences in marginal costs of balance sheet for

⁹ Formally, we need an assumption ensuring that marginal cost advantages (ε 's) are not swamped by differences in adjustment costs (c 's) or outside demand (through the V 's).

group I and $\sim I$ shrink but are not fully equated, the correlation of spreads will rise but not all the way to 1. We further explore arbitrage dynamics within the model in [Internet Appendix Section III](#).¹⁰

E. Empirical Implications

The model highlights what we can learn from spreads alone and what conclusions require ancillary data. For instance, a high-dimensional factor structure for spreads rejects simple models in which both balance-sheet and funding integration obtain and the representative intermediary is subject to a single constraint. As equation (3) shows, however, a high-dimensional factor structure in and of itself does not distinguish between situations in which (i) both balance-sheet and funding integration obtain but the representative intermediary is subject to many constraints and (ii) either balance sheets or funding markets are segmented. As equation (4) shows, the empirical signature of funding segmentation is a covariance between certain spreads and certain funding rates. Similarly, equation (6) shows that the empirical signature of balance-sheet segmentation is a covariance between certain spreads and individual intermediary balance-sheet costs. Our empirical analysis below follows this outline. We begin by describing the factor structure of arbitrage spreads and then provide direct evidence of both types of segmentation. For instance, we directly show that certain spreads comove with the costs of particular types of funding. We also show that certain spreads respond directly to shocks to the balance sheets of specific intermediaries.

Finally, while we have discussed funding and balance-sheet segmentation separately, it is worth noting that they are not mutually exclusive. We think of segmented funding and segmented balance sheets simultaneously driving individual arbitrage spreads. For instance, below we show that equity spot-futures spreads depend both on the TED spread, which captures conditions in the market for unsecured funding, and JPMorgan's balance-sheet constraints. In other words, JPMorgan is acting as an arbitrageur in these trades, so its balance-sheet conditions impact these trades, and is raising financing from the frictional unsecured funding market to finance them. Moreover, the dealer banks sometimes finance arbitrage and sometimes participate directly as arbitrageurs, depending on the type of trade. Our results below broadly suggest that dealer banks likely participate as arbitrageurs for unsecured arbitrages. In contrast, for secured arbitrages, dealers likely finance arbitrage trading through their role in the repo market (and as prime brokers).

II. The Factor Structure of Arbitrage

A. Data

Our main analysis sample covers 32 arbitrage trades over the period from January 1, 2010 to February 29, 2020. This period spans the post-GFC era

¹⁰ The [Internet Appendix](#) may be found in the online version of this article.

and predates the Covid-19 pandemic.¹¹ For each arbitrage trade, we construct an implied riskless rate based on observed asset prices and then subtract a maturity-matched benchmark riskless rate. For arbitrage trades that mature in less than two years, the benchmark is based on overnight indexed swap (OIS) rates; for longer-maturity trades, it is based on Treasury yields. Our choice of benchmark rates means that our arbitrage spreads do not represent true riskless profits that are available to unconstrained intermediaries, since they are not constructed using the exact funding rate that a trader implementing the arbitrage would face. Instead, our arbitrage spreads capture funding and other frictions faced by arbitrageurs, which are precisely what we seek to characterize. A detailed description of each arbitrage spread and its construction is provided in [Internet Appendix Section I](#). Here, we provide a short description of the trades, which can be grouped into seven broad categories, or “strategies.” The code for constructing arbitrage spreads is in a publicly available Github repository.¹²

A.1. Arbitrage Spreads

Foreign Exchange Arbitrage: We follow Du, Tepper, and Verdelhan (2018) and measure arbitrage spreads in foreign exchange markets with deviations from CIP. For each currency we study, we define the CIP arbitrage spread as the difference between the dollar OIS rate and a synthetic riskless rate that is implied by currency forwards, currency spot rates, and foreign OIS rates. We build CIP arbitrage spreads for all G-10 currencies except the Danish and Norwegian kroner because OIS rates are not available for these two currencies. We use three-month CIP violations to avoid any confounding effects that the quarter-end spikes documented in Du, Tepper, and Verdelhan (2018) may have on correlations. We obtain data on spot and forward exchange rates and OIS rates from Bloomberg. See [Internet Appendix Section A](#) for more details.

Equity Options Arbitrage (Box Arbitrage): We infer riskless rates and arbitrage spreads from S&P 500 (SPX) equity options based on the put-call parity relationship. As discussed in Ronn and Ronn (1989) and van Binsbergen, Diamond, and Grotteria (2019), implied riskless rates from put-call parity are often called box rates in practice. We adopt this naming convention and refer to this arbitrage as the box trade in the remainder of the paper. We take box rates for 6-, 12-, and 18-month tenors directly from van Binsbergen, Diamond, and Grotteria (2019), who estimate them using minute-by-minute pricing data for SPX options through 2018. We next follow van Binsbergen, Diamond, and Grotteria (2019)’s methodology to extend the data to 2020 using SPX option data purchased directly from the Chicago Board Options Exchange (CBOE).

¹¹ In [Internet Appendix Section B](#), we further study the behavior of arbitrage spreads during the Covid-19 pandemic and before the Dodd-Frank era and show that correlations were low before the GFC and at the onset of the pandemic.

¹² See <https://github.com/esiriwardane/arbitrage-spreads-public>.

Arbitrage spreads are then computed by subtracting off a maturity-matched OIS rate.

Equity Spot-Futures Arbitrage: For equity futures markets, we measure arbitrage spreads based on violations of spot-futures parity. As we discuss in [Internet Appendix Section C](#), the spot and futures markets for equities close at different times, which prevents us from using the spot-futures parity relationship to accurately compute implied riskless rates from closing prices alone.¹³ Instead, we compute implied forward rates based on the relative pricing of futures contracts with different tenors. To illustrate, consider a futures contract on an asset that does not pay a dividend. In this case, spot-futures parity implies that the current futures price F_{T_1} for a contract with tenor T_1 and the spot price S satisfy $F_{T_1} = S(1 + r_{T_1})$, where r_{T_1} is the riskless rate between today and T_1 . Next, consider another futures contract with tenor $T_2 > T_1$. Under the parity condition, the ratio of the two futures prices F_{T_2}/F_{T_1} equals the gross forward rate $1 + f_{T_1, T_2}$ between T_1 and T_2 .

We estimate implied forward rates using Bloomberg futures data on the SPX, Dow Jones Industrial, and Nasdaq 100 indices. For each index, our analysis is based on the nearby and first-deferred contracts, which are the most liquid. [Internet Appendix Section C](#) provides more details about our implementation, including how we account for dividends and compute arbitrage spreads from implied forward rates.

Treasury Spot-Futures Arbitrage: For Treasury futures markets, we measure arbitrage spreads based on violations of spot-futures parity, following Fleckenstein and Longstaff (2020) and Barth and Kahn (2021). We study five such trades, associated with the first-deferred futures contract on the 2-, 5-, 10-, 20-, and 30-year Treasury. We measure arbitrage spreads using the first-deferred contract to avoid complications with the nearby contract in the futures delivery month (Fleckenstein and Longstaff (2020)). We obtain futures-implied riskless rates directly from Bloomberg and define arbitrage spreads by subtracting off a maturity-matched OIS rate. See [Internet Appendix Section D](#) for more details.

Treasury Swap Arbitrage: For interest rate swap markets, we measure arbitrage spreads using OIS swap spreads, defined as the difference between the fixed rate on overnight indexed swaps and Treasury yields. We study seven such trades, associated with 1-, 2-, 3-, 5-, 10-, 20-, and 30-year Treasuries. OIS swap rates are from Bloomberg. As discussed in Jermann (2020), Du, Hébert, and Li (2022), and Hanson, Malkhozov, and Venter (2022a), only negative OIS swap spreads indicate a guaranteed arbitrage. In [Internet Appendix Section E](#), we show that this condition is satisfied for the large majority of observations in our analysis sample.

¹³ Hazelkorn, Moskowitz, and Vasudevan (2023) avoid this issue by using high-frequency data for both futures and spot markets.

TIPS-Treasury Arbitrage: We build on Fleckenstein, Longstaff, and Lustig (2014) and construct the difference in yield between a synthetic nominal Treasury, constructed using TIPS and inflation swaps, and the true nominal Treasury yield. Unlike Fleckenstein, Longstaff, and Lustig (2014), who construct the arbitrage at the security level, we use the zero-coupon constant-maturity TIPS and Treasury yields that are published by the Federal Reserve based on Gürkaynak, Sack, and Wright (2007, 2010).¹⁴ Constant-maturity inflation swap data are taken from Bloomberg. In Internet Appendix Section F, we provide additional details and compare our series to that constructed in Fleckenstein, Longstaff, and Lustig (2014). We focus on 2-, 5-, 10-, and 20-year maturities to ensure the underlying TIPS securities used to construct the arbitrages are sufficiently liquid.

CDS-Bond Arbitrage: For U.S. corporate bond and CDS markets, we follow Duffie (1999) and measure arbitrage spreads based on the difference between cash-bond implied credit spreads and CDS spreads. Cash bond and CDS pricing data both come from Markit. We form CDS-bond bases for both investment-grade and high-yield bonds, aggregating over bonds in each ratings category. The average number of bonds used to compute the daily investment-grade and high-yield bases is 1,690 and 307, respectively. Internet Appendix Section G contains the full construction methodology.

Summary Statistics: Table I provides summary statistics. The data are daily and spreads are reported in annualized basis points (bps). Unless otherwise noted, we work with absolute values of spreads since the sign of the spread depends on whether arbitrageurs are net long or short a particular leg of the trade. The number of observations varies slightly across trades, mainly due to availability from raw data providers (e.g., Bloomberg vs. Markit) and differences in trading holidays across swaps and futures markets.

Table I shows that there is significant variation in spreads, both across trades on average and within trades over time. For many individual trades, the daily standard deviation of spreads is around half the mean spread. Figure 1 helps visualize this variation by plotting average spreads by broad strategy, at both the daily and monthly frequencies. Table I also indicates that spreads are highly persistent at the daily level, a fact that we are careful to account for when conducting statistical inference.

A.2. Quantity Data

In addition to data on arbitrage spreads, we use data from the Commodity Futures Trading Commission (CFTC) on quantities.¹⁵ The CFTC publishes weekly “Traders in Financial Futures; Futures-and-Options Combined” reports, which break down open interest for futures and options markets in

¹⁴ The yield curve models can be found at <https://www.federalreserve.gov/data/yield-curve-models.htm>.

¹⁵ The data are available at <https://www.cftc.gov/MarketReports/CommitmentsofTraders/index.htm>.

Table I
Summary Statistics for Arbitrage Spreads

This table presents summary statistics on the absolute values of different arbitrage spreads. Trades are grouped by strategy (e.g., CIP). All CIP trades are for three-month tenors. SPX, DJX, and NDAQ SF are spot-futures arbitrages for the S&P 500, Dow Jones, and Nasdaq indices, respectively. Treasury i Y SF is the Treasury spot-futures arbitrage for i -year maturity Treasuries. CDS-bond denotes the average CDS-bond basis for investment-grade (IG) and high-yield (HY) firms. See Section II.A and Internet Appendix Section I for details on the construction of arbitrage trades. The column AR1 is the coefficient from an AR(1) model estimated from daily data. The columns First and Last are the month and year of the first and last observation for each series.

	Mean	p50	SD	Min	Max	AR1	First	Last	N
Box 6m	35	35	11	0	82	0.897	Jan-10	Feb-20	2,534
Box 12m	34	34	10	0	87	0.914	Jan-10	Feb-20	2,534
Box 18m	33	32	9	2	64	0.943	Jan-10	Feb-20	2,534
Dow SF	45	45	24	0	139	0.976	Jan-10	Feb-20	2,497
NDAQ SF	38	39	22	0	123	0.978	Jan-10	Feb-20	2,496
SPX SF	42	41	20	0	116	0.982	Jan-10	Feb-20	2,495
AUD CIP	13	11	10	0	59	0.933	Jan-10	Feb-20	2,541
CAD CIP	12	9	10	0	62	0.983	Jan-10	Feb-20	2,539
CHF CIP	51	47	27	11	198	0.974	Jan-10	Feb-20	2,541
EUR CIP	35	33	21	0	118	0.986	Jan-10	Feb-20	2,540
GPB CIP	17	12	14	0	93	0.986	Jan-10	Feb-20	2,540
JPY CIP	45	41	24	10	125	0.986	Jan-10	Feb-20	2,541
NZD CIP	11	11	6	0	36	0.873	Jan-10	Feb-20	2,540
SEK CIP	27	21	22	0	99	0.985	Jan-10	Feb-20	2,541
Treasury 2Y SF	13	12	9	0	62	0.930	Jan-10	Feb-20	2,347
Treasury 5Y SF	12	9	11	0	58	0.953	Jan-10	Feb-20	2,388
Treasury 10Y SF	18	15	15	0	113	0.930	Jan-10	Feb-20	2,477
Treasury 20Y SF	17	13	14	0	79	0.898	Jan-10	Feb-20	2,494
Treasury 30Y SF	11	9	10	0	180	0.655	Jan-10	Feb-20	1,734
Treasury-Swap 1Y	6	5	5	0	32	0.964	Jan-10	Feb-20	2,541
Treasury-Swap 2Y	10	9	6	0	34	0.964	Jan-10	Feb-20	2,541
Treasury-Swap 3Y	12	10	8	0	36	0.982	Jan-10	Feb-20	2,541
Treasury-Swap 5Y	17	15	10	0	44	0.984	Jan-10	Feb-20	2,541
Treasury-Swap 10Y	26	25	12	0	59	0.986	Jan-10	Feb-20	2,541
Treasury-Swap 20Y	35	35	15	8	70	0.990	Sep-11	Feb-20	2,105
Treasury-Swap 30Y	54	51	19	23	100	0.995	Sep-11	Feb-20	2,105
TIPS-Treasury 2Y	21	23	12	0	54	0.968	Jan-10	Feb-20	2,541
TIPS-Treasury 5Y	19	20	8	0	56	0.981	Jan-10	Feb-20	2,541
TIPS-Treasury 10Y	25	25	7	1	40	0.976	Jan-10	Feb-20	2,541
TIPS-Treasury 20Y	26	26	8	1	47	0.974	Jan-10	Feb-20	2,541
CDS-Bond IG	22	20	13	0	79	0.973	Jan-10	Feb-20	2,540
CDS-Bond HY	65	59	36	1	188	0.989	Jan-10	Feb-20	2,540

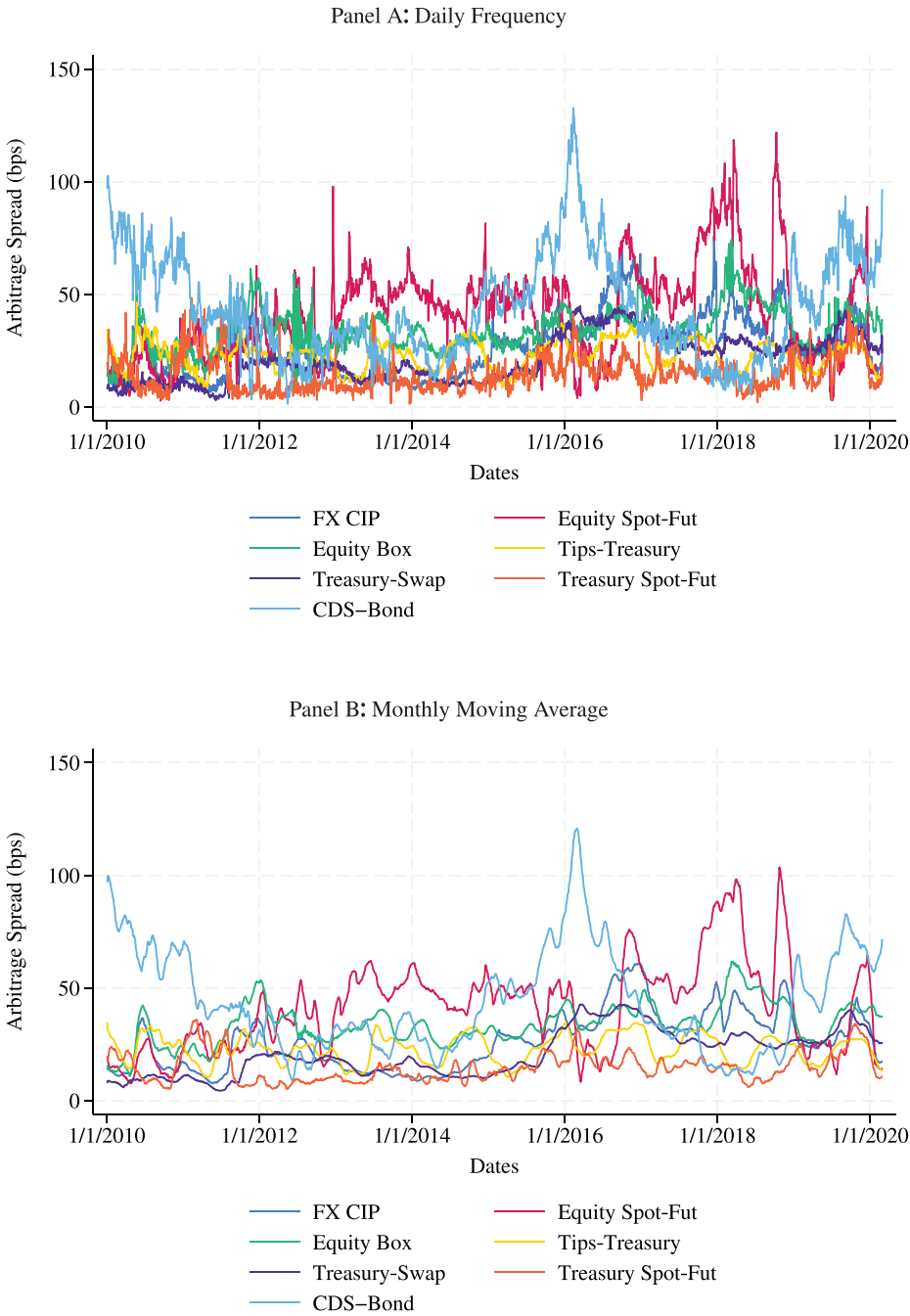


Figure 1. Strategy-level average arbitrage. Panel A shows the average arbitrage spread by strategy at the daily frequency. Panel B plots a monthly (22 trading days) moving average of the daily series for each strategy. (Color figure can be viewed at [wileyonlinelibrary.com](#))

which 20 or more traders hold large positions. The position data are supplied by clearinghouses and other reporting firms. The reports separate positions into four trader types: dealers, asset managers, leveraged funds, and other reporting entities. These classifications are based on the predominant business purpose self-reported by traders on CFTC Form 40. These data are weekly and begin in July 2010.

A.3. Money Market Fund Holdings

We obtain data on the holdings and total net assets (TNAs) of MMFs from Crane data. The data are compiled from form N-MFP, which MMFs are required to file with the Securities and Exchange Commission (SEC) every month.

A.4. Hedge Fund Returns

We also use hedge fund returns from the Prequin Pro Hedge Fund Database. This database includes performance data on over 24,000 hedge funds. Importantly for our purposes, the database contains descriptive information on fund strategies, which allows us to focus on funds that self-report being involved in the arbitrage trades that we study.

B. Characterizing Arbitrage Comovement

B.1. Baseline Results

We now turn to our first main result, namely, that the correlation between arbitrage spreads is low. Figure 2 presents this result graphically, depicting a heat map of pairwise correlations between the absolute value of different spreads. Darker red indicates higher positive correlations. With the exception of the diagonal, little of the figure is dark red, indicating that correlations are generally low.

Table II, Panel A provides formal statistical evidence on pairwise correlations. The average pairwise correlation is 0.22, and the 75th percentile of pairwise correlations is 0.42. These results are at odds with simple structures for the intermediary sector in which there is only a single balance-sheet constraint or a single funding factor. As shown in Section I, in these cases arbitrage spreads should be perfectly correlated. In Figure 3, we conduct a principal component analysis (PCA) of spreads. Consistent with the low correlations documented above, it takes 11 PCs to cumulatively explain 90% of the variation in arbitrage spreads. Furthermore, the last column of Table II, Panel A shows that we can reject the null of equal correlations across all arbitrage pairs.¹⁶ Thus, the data suggest a complex structure for the intermediary sector.

¹⁶ The low daily correlation between arbitrage spreads is more striking given that the high daily persistence of spreads (Table I) should, if anything, generate spurious relationships between arbitrages (Granger and Newbold (1974)). The average pairwise correlation of changes in arbitrage spreads is similarly low at 5%.

Table II
Correlations within and across Arbitrage Strategies

Each panel of this table summarizes the distribution of pairwise correlations for a different subsample of the data. Panel A contains all arbitrage trades. Panel B contains all arbitrage trades with tenors of less than six months. Panel C contains all arbitrage trades with tenors greater than six months and less than or equal to three years. Panel D contains all arbitrage trades with tenor greater than three years, including the CDS-bond basis strategies. Panel E is for the subsample of dates on which each arbitrage in a given pair exceeds the average six-month CDS spread for the set of dealers who have been designated as primary dealers by the Federal Reserve Bank of New York since 2010. The columns under p -value report tests of the null that the average pairwise correlation is above 0.67 and the null that all pairwise correlations are equal.

Panel A: Distribution of All Pairwise Correlations

ρ_{ij}								p -Value	
Mean	SD	Min	p25	p50	p75	Max	N	$\bar{\rho} > 0.67$	$\rho_{ij} = \rho$
0.22	0.30	-0.55	0.00	0.17	0.42	0.96	496	0.00	0.00

90% of pairs reject H_0 : $\rho_{ij} > 0.67$

Panel B: Distribution of Pairwise Correlations for Short-Tenor Trades

ρ_{ij}								p -Value	
Mean	SD	Min	p25	p50	p75	Max	N	$\bar{\rho} > 0.67$	$\rho_{ij} = \rho$
0.19	0.32	-0.41	-0.02	0.15	0.35	0.89	120	0.00	0.00

87% of pairs reject H_0 : $\rho_{ij} > 0.67$

Panel C: Distribution of Pairwise Correlations for Medium-Tenor Trades

ρ_{ij}								p -Value	
Mean	SD	Min	p25	p50	p75	Max	N	$\bar{\rho} > 0.67$	$\rho_{ij} = \rho$
0.39	0.28	0.02	0.21	0.32	0.50	0.93	21	0.00	0.00

81% of pairs reject H_0 : $\rho_{ij} > 0.67$

Panel D: Distribution of Pairwise Correlations for Long-Tenor Trades

ρ_{ij}								p -Value	
Mean	SD	Min	p25	p50	p75	Max	N	$\bar{\rho} > 0.67$	$\rho_{ij} = \rho$
0.37	0.30	-0.22	0.13	0.32	0.53	0.96	36	0.00	0.00

81% of pairs reject H_0 : $\rho_{ij} > 0.67$

(Continued)

Table II—Continued

Panel E: Conditional on Dealer Credit Spreads								p-Value	
ρ_{ij}									
Mean	SD	Min	p25	p50	p75	Max	N	$\bar{\rho} > 0.67$	$\rho_{ij} = \rho$
0.27	0.28	−0.48	0.08	0.27	0.46	0.97	492	0.00	0.00

86% of pairs reject H_0 : $\rho_{ij} > 0.67$

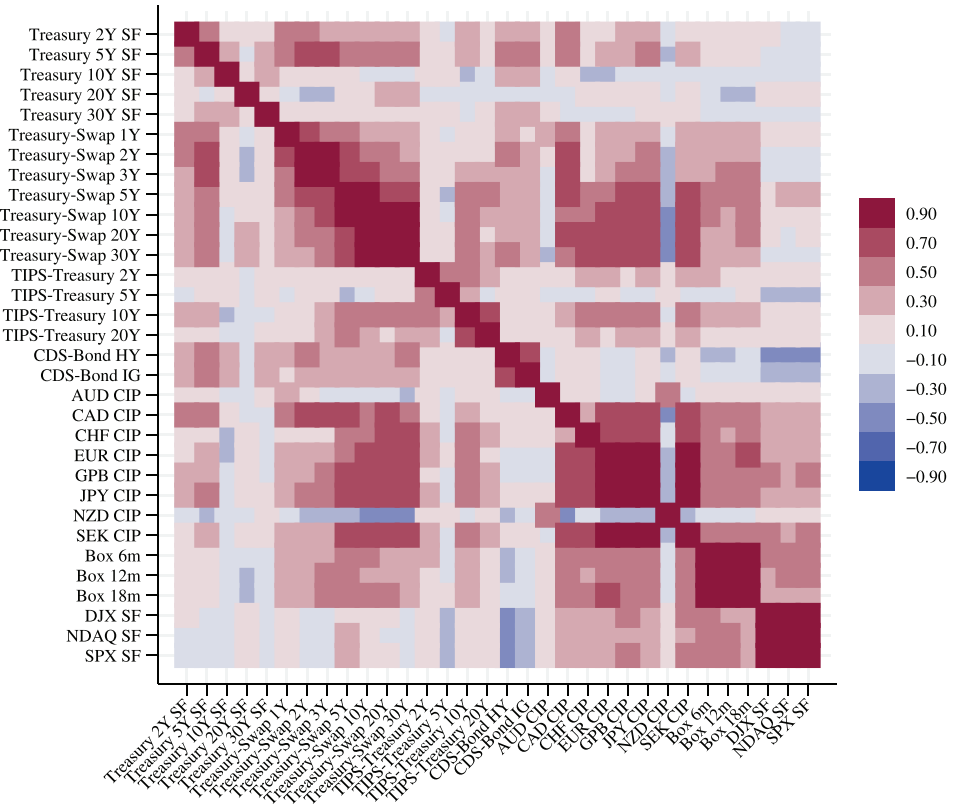


Figure 2. Correlation of arbitrage spreads. The figure shows the pairwise correlation matrix of the absolute value of arbitrage spreads across all trades in our sample. See Section II.A for details on each trade. Data are daily and span January 1, 2010 to February 29, 2020. (Color figure can be viewed at [wileyonlinelibrary.com](#))

Either balance-sheet and funding integration hold but the representative intermediary faces a large number of different constraints, or there is significant segmentation in arbitrage.

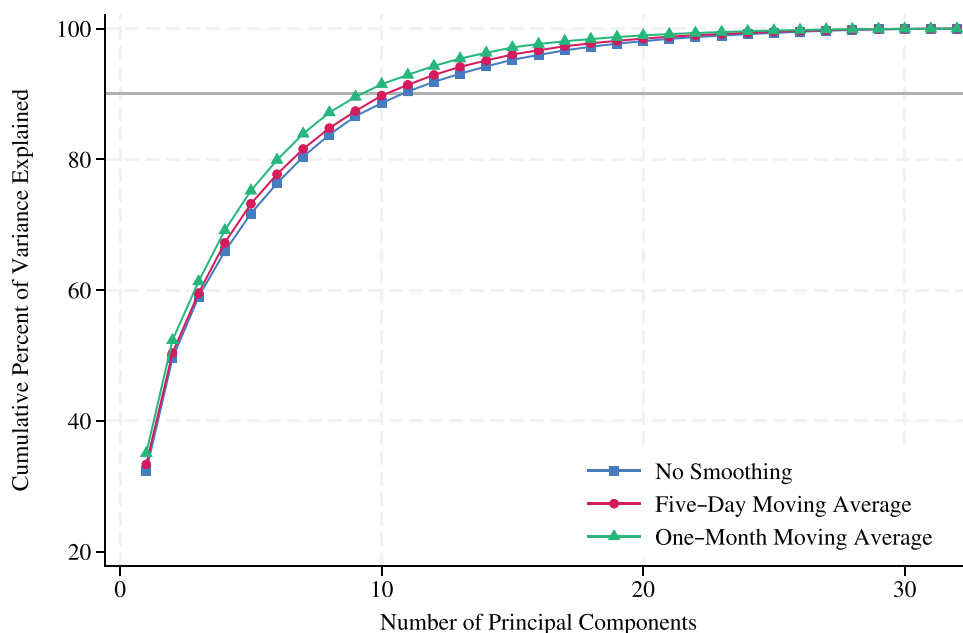


Figure 3. The factor structure of arbitrage spreads. This figure summarizes principal component analysis for the absolute values of arbitrage spreads in our sample. Each line shows the results of principal component analysis after we smooth our arbitrage spreads over a different moving-average window. The *x*-axis shows the number of components and the *y*-axis shows the cumulative proportion of variance captured by those components. The gray horizontal line on the plot is at the 90% level. See Section II.A for details on each trade. Data are daily and span January 1, 2010 to February 29, 2020. (Color figure can be viewed at wileyonlinelibrary.com)

B.2. Measurement Error

Measurement error is an important issue to consider when interpreting low arbitrage correlations. If the observed spread is the true spread plus noise, then the correlation between observed spreads will suffer from attenuation bias, understating the correlation between true spreads.

We address potential measurement error in a few complementary ways. First, Table II, Panel A shows that the average pairwise correlation is significantly less than one. We reject the null that the average pairwise correlation of all spreads is greater than 0.67, suggesting that the variance of any measurement error would have to be quite large to account for our results. Second, smoothing should increase the ratio of variation driven by true spreads as opposed to noise, but it has little effect on our results. Figure 3 shows that we obtain very similar results if we compute PCs after taking a five-day or one-month moving average of spreads. Even after taking one-month moving averages of spreads, it takes 10 PCs to cumulatively explain over 90% of the variation in our arbitrage spreads. Third, in Section A of the

[Internet Appendix](#), we directly adjust for attenuation bias using instrumental variables and show that correlations remain low even after this adjustment.

B.3. The Term Structure of Arbitrage

The term structure is another important consideration when interpreting the correlation of arbitrage spreads. This topic has been studied in recent work by Du, Hébert, and Li (2022) and Hanson, Malkhozov, and Venter (2022a). In our setting, intertemporal considerations may change the correlation structure of arbitrage even with an integrated intermediary sector. For instance, trades that differ in tenor may not be perfectly correlated because long-tenor trades reflect expected future balance-sheet costs while short-tenor trades reflect only current balance-sheet costs. [Internet Appendix Section III](#) modifies our baseline model to formalize these considerations; for a fuller treatment, see Du, Hébert, and Li (2022) and Hanson, Malkhozov, and Venter (2022a).

A simple way to mitigate these concerns is to compare trades with similar tenors. Table II implements this idea by grouping trades into one of three buckets based on their tenor. The short-tenor bucket contains all trades whose maturity is always less than six months, namely, CIP, equity spot-futures, and Treasury-spot futures. Medium-tenor trades are those whose tenor is greater than or equal to six months but less than or equal to three years. Finally, long-tenor trades are those with a tenor exceeding three years. The CDS-bond arbitrages are included with long-tenor trades based on the typical tenor of their underlying bonds.

Table II shows the distribution of pairwise correlations for short-, medium-, and long-tenor trades. The broad takeaway from the table is that correlations remain relatively low, even within trades of the same tenor. For example, in Table II, Panel B, the average correlation among short-tenor trades is 0.19 and the 75th percentile of correlations is 0.35. Because short-tenor trades are less exposed to mark-to-market risk, the low correlation among these trades cuts strongly against conventional models featuring an integrated intermediary sector. As a more extreme example, consider the correlation between two *overnight* arbitrages: (i) the spread between interest paid on excess reserves (IOER) and the effective federal funds rate (Banegas and Tase (2020)), and (ii) the spread between the general collateral repo rate and the tri-party repo rate (TPR) (Correa, Du, and Gordon (2020)). Although these trades have virtually no mark-to-market risk, their correlation during our analysis sample is 23%, a clear indication of segmentation.¹⁷

¹⁷ We thank Wenxin Du for bringing this example to our attention. To construct the IOER arbitrage, we use the first percentile of the overnight federal funds rate from the New York Federal Reserve. This necessarily focuses on the depository institutions that can earn an arbitrage profit because they are able to borrow in the Federal funds market at favorable rates, most likely large banks (Banegas and Tase (2020)). The correlation between the General Collateral Finance repo rate (GCF)-TPR and the volume-weighted IOER arbitrage spread (truncated at zero) is 21%. TPRs from BNY Mellon are available starting in 2016.

Tables II, Panels C and D summarize the distribution of correlations for medium- and long-tenor trades, respectively. The degree of integration appears higher in these trades compared to those with short tenors, at least as evidenced by their larger average pairwise correlation (roughly 40% vs. 20%). This could occur if some institutions specialize in longer-tenor trades, perhaps because they are better equipped to manage mark-to-market risk. For example, access to stable funding may allow larger dealers to mitigate noise-trader risk because they can more reliably roll over repo. Specialization could also arise because longer-tenor arbitrages use fixed-income derivatives like interest rate swaps and CDSs. With that said, the more important point is that correlations within medium- and long-tenor trades are still far from perfect. In both groups, the average correlation is below 40% and the 75th percentile is around 50%. These relatively low correlations further support our main argument that arbitrage is more segmented than previously acknowledged in the literature.

B.4. Zones of Arbitrageur Inaction

Arbitrageurs may not find it profitable to enter a market unless the level of arbitrage is sufficiently high due to debt overhang issues—as argued by Andersen, Duffie, and Song (2019), an intermediary facing a debt overhang problem will only enter a riskless arbitrage if the return is higher than its credit spread. Consequently, small arbitrage spreads may appear uncorrelated because of this “zone of inaction” even if the intermediary sector is integrated. To address this concern, we first compute the cross-sectional average six-month CDS spread across all New York Federal Reserve primary dealers, denoted d_t .¹⁸ Next, we compute the pairwise correlation of each pair of trades, conditional on both exceeding d_t . The logic of Andersen, Duffie, and Song (2019) suggests that dealers should be active when arbitrage spreads exceed this threshold. Table II, Panel E summarizes the distribution of pairwise correlations when arbitrage spreads exceed dealer credit spreads. The average correlation remains below 30% and the distribution is comparable to that observed in the full sample (Table II, Panel A).

C. Supply versus Demand

As shown in Section I, once we depart from the integrated intermediary benchmark, the correlations of arbitrage spreads are determined by both arbitrageur supply and end-user demand. One might wonder whether the low correlations that we document arise primarily from the supply side or the demand side. To shed light on this question, we follow the large literature on sign-restricted SVARs (Uhlig (2005); Arias, Rubio-Ramírez, and Waggoner (2018)). While this approach was originally developed for macroeconomic time-series analysis, it has been applied more recently in financial markets by Goldberg and Nozawa (2021) and Hanson, Malkhozov, and Venter (2022b), among others.

¹⁸ We include any dealer that has been listed as a primary dealer since 2010.

Denote $Y_t = [s_t \ q_t]'$ as the vector of arbitrage spreads and quantities for a given trade. Quantities are defined more precisely below, but for now it is sufficient to think of them as the size of the derivative market that supports a particular arbitrage trade (e.g., gross outstanding futures). The dynamics of Y_t are assumed to take the form

$$BY_t = A_0 + A_1 Y_{t-1} + \varepsilon_t, \quad (8)$$

where $\varepsilon_t = [\varepsilon_{s,t} \ \varepsilon_{d,t}]'$ are primitive orthonormal shocks. For reasons discussed below, the structural shock $\varepsilon_{s,t}$ can be thought of as a supply shock that emanates from arbitrageurs, whereas $\varepsilon_{d,t}$ can be thought of as demand for the derivatives relative to cash securities arising from end-users like insurance or pension funds. Assuming B is invertible, the reduced-form VAR implied by the structural model in (8) is

$$Y_t = \Phi_0 + \Phi_1 Y_{t-1} + u_t,$$

where $\Phi_0 = B^{-1}A_0$ and $\Phi_1 = B^{-1}A_1$. The covariance matrix of the reduced-form residuals $u_t = B^{-1}\varepsilon_t = [u_{s,t} \ u_{q,t}]'$ is given by Σ_u and depends only on the matrix B . While the reduced-form parameters (Φ_0, Φ_1, Σ_u) can be estimated by ordinary least squares (OLS) or Bayesian methods, the structural parameters (A_0, A_1, B) cannot be identified without imposing additional restrictions. One solution to this problem is to impose restrictions on the sign of the impact of structural shocks on the reduced-form shocks. In our context, we assume the following sign restrictions:

$$\begin{bmatrix} u_{s,t} \\ u_{q,t} \end{bmatrix} = \underbrace{\begin{bmatrix} - & + \\ + & + \end{bmatrix}}_{B^{-1}} \begin{bmatrix} \varepsilon_{s,t} \\ \varepsilon_{d,t} \end{bmatrix}. \quad (9)$$

These restrictions make our labeling of the structural shocks more clear. We interpret $\varepsilon_{s,t}$ as a supply shock because it lowers arbitrage spreads and raises quantities. Conversely, we interpret $\varepsilon_{d,t}$ as a demand shock because it raises both arbitrage spreads and quantities.

We estimate the sign-restricted SVAR using weekly data separately for each futures-based arbitrage trade, as these are the only trades for which quantity data are publicly available. The quantity for a given trade is defined as the gross amount of positions held by all reporting traders, including spread trades, in the CFTC's weekly "Traders in Financial Futures; Futures-and-Options Combined Reports," described in more detail in Section II.A.2. Arbitrage spreads equal the absolute value of raw spreads because this is the most natural object on which to impose the sign restrictions.¹⁹

¹⁹ The 30-year Treasury spot-futures arbitrage is excluded because it is more prone to missing arbitrage spread data (see Table I). For the remaining series, missing arbitrage spreads are filled with their last nonmissing value.

Our estimation strategy closely follows Arias, Rubio-Ramírez, and Waggoner (2018), who develop an efficient algorithm to estimate sign-restricted SVARs with appropriately conservative confidence intervals. First, we employ Bayesian methods to estimate the reduced-form VAR parameters under a Normal-Wishart prior. Next, we randomly draw from the posterior distribution of the reduced-form parameters and use the Cholesky decomposition of Σ_u to obtain a B that satisfies the sign restriction in (9).²⁰ This process is repeated 1,000 times to generate a set of structural parameters that all satisfy the sign restriction, reflecting the well-known result that sign-restricted SVARs are generally only set-identified. With some abuse of notation, let Θ_i be the set of identified structural parameters when estimating the model for trade i . Each element of this set implies its own sequence of supply and demand shocks, and we denote $(\varepsilon_{i,t}^s, \varepsilon_{i,t}^d)$ as the shocks associated with the median parameter draw from Θ_i .

Note that since the model is estimated separately for each trade, there are no restrictions on the covariance of supply and demand shocks *across* trades. Figure 4(A) depicts the correlation of estimated supply shocks $\varepsilon_{i,t}^s$ across trades. The plot shows that supply shocks are far from perfectly correlated, as would be the case if arbitrage were perfectly integrated. The average pairwise correlation of 16% is low and supports our broad argument that supply-side frictions like balance-sheet or funding segmentation have a first-order impact on asset prices. With that said, Figure 4(A) also reveals portions of the market where arbitrage appears more integrated. For example, perhaps unsurprisingly, the average correlation of supply shocks within equity-spot futures arbitrage is higher at 62%.²¹

Figure 4(B) depicts the correlation of the demand shocks $\varepsilon_{i,t}^d$ from the SVAR. Again, these shocks are most naturally interpreted as the relative demand for futures versus spot contracts from end-users like insurance companies and pension funds. Consistent with the intuition that demand is also segmented, the average correlation of the demand shocks equals 16%. However, there are again portions of the capital markets where demand shocks are more correlated. For instance, within equity spot-futures trades, the average correlation of demand shocks equals 64%.

In Internet Appendix Section C, we use the full distribution of shocks to put confidence bands on the correlation. We find that the average across trades of the 1% upper bound on the pairwise correlation of supply shocks equals 24%,

²⁰ Our exact implementation is based on the open source VAR Matlab toolbox of Breitenlechner, Geiger, and Sindermann (2019), found at <https://eeecon.uibk.ac.at/~breitenlechner/data/ZeroSignVAR.zip>.

²¹ In contrast, the correlation of supply shocks within Treasury futures-arbitrage appears to be quite low. This low correlation appears to be due to both prices and quantities. The average correlation of spreads across Treasury-futures arbitrage is low at 20%. The correlation of quantities is higher at 70%, but both contribute to the lower correlation of supply shocks that we document. Our analysis does not directly speak to whether these low correlations are driven by balance-sheet segmentation or funding segmentation, but they are consistent with the literature on preferred habitats within the Treasury market (Vayanos and Vila (2021)).

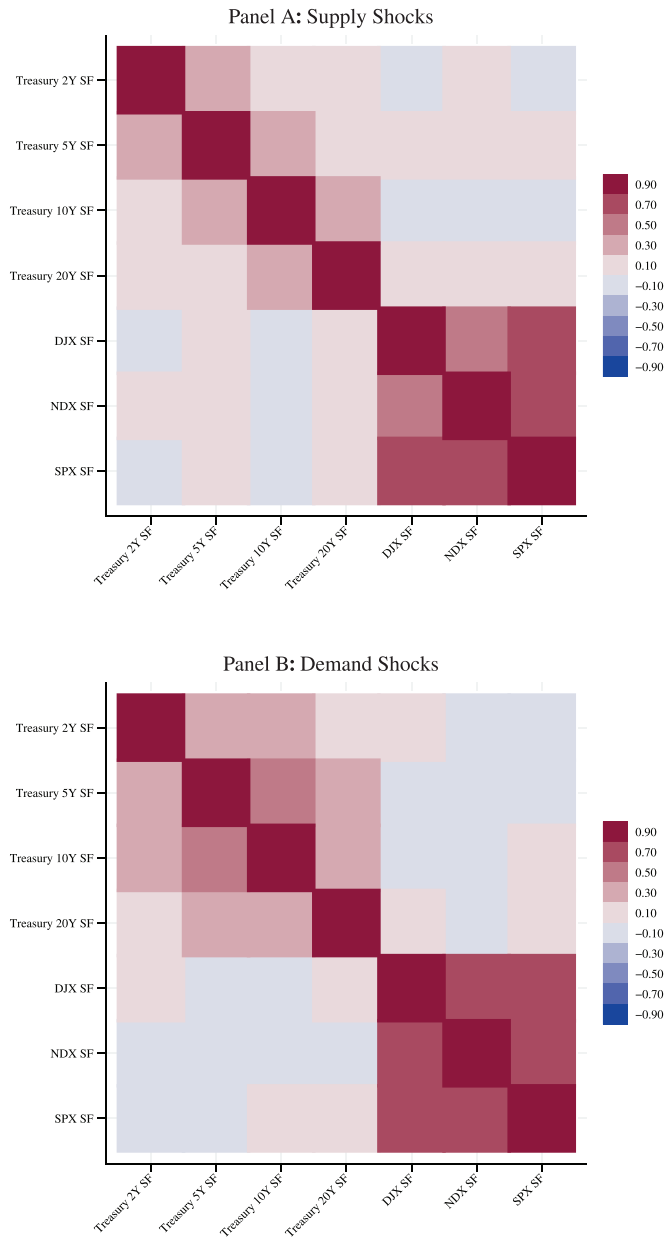


Figure 4. Correlation of arbitrage supply and demand shocks. This figure shows the correlation of supply (Panel A) and demand (Panel B) shocks based on a sign-restricted SVAR that is estimated separately for each trade. For a given trade i , we first estimate a sign-restricted SVAR with one lag following Uhlig (2005) and Arias, Rubio-Ramírez, and Waggoner (2018). Using the median estimated parameter set, we then construct the implied supply and demand shock series. The top and bottom panels show the correlation matrix of the respective supply and demand shock series across trades. Data are weekly from 2010 to 2020. See Section II.C for more details. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.com))

suggesting that arbitrage is far from integrated for futures-based trades with 99% confidence. In [Internet Appendix Section D](#), we use the SVAR to further decompose the covariance between a pair of arbitrage spreads into three terms suggested by the model in [Section I](#): (i) covariance between supply shocks, (ii) covariance between demand shocks, and (iii) covariance between supply and demand shocks across two trades. For the average pair, comovement between supply shocks contributes 14% to the overall covariance, and comovement between demand shocks contributes 26%. Thus, according to the SVAR, comovement between supply and demand shocks contribute similarly to the overall observed covariance of arbitrage spreads. The remaining covariance is attributable to cross-market supply and demand terms.

III. Segmented Funding

In this section, we focus on funding frictions as a key driver of the low correlations of supply shocks documented above. As discussed in [Section I](#), the underlying violation of the Modigliani and Miller (1958) theorem is that certain riskless portfolios cannot be funded at the riskless rate. For instance, the equity spot-futures arbitrage involves holding the underlying equities and selling equity futures. Taken together, this position is riskless, but it cannot be funded with, for instance, Treasury repo. As the cost of funding for certain arbitrage trades moves, spreads move as well.

We proceed in three steps. We start with suggestive evidence that there are differences in funding structures across the different arbitrage strategies we study. We then provide more formal empirical evidence that movements in funding costs affect arbitrage spreads. In particular, we show that they help explain the relatively high degree of comovement between the box, CIP, and equity-spot futures spreads in [Table I](#) of the [Internet Appendix](#), and the relatively low degree of comovement between those spreads and the others we study. Finally, we show that specialization in funding creates segmentation that goes beyond the divide between unsecured and secured funding markets.

A. Differences in Funding Structures

There are meaningful differences in the availability of secured financing across arbitrage strategies. In [Internet Appendix Section E](#), we show that the Treasury spot-futures, Treasury-swap, and TIPS-Treasury arbitrages can be largely financed with Treasury repo, requiring only a 2% margin. In other words, intermediaries need little unsecured debt or equity funding to enter into these arbitrages. Conversely, the box, CIP, and equity spot-futures arbitrages require higher margins between 8% and 12%. For these arbitrages, unsecured funding conditions are much more important. We therefore frequently group these trades together, labeling them “unsecured,” while we label the remaining trades (Treasury spot-futures, Treasury-swap, TIPS-Treasury, and CDS-bond) “secured.”

Table III
Arbitrage-Implied Riskless Rates and Funding Conditions

This table shows monthly OLS regressions of changes in arbitrage-implied riskless rates on changes in maturity-matched Treasury yields and TED spreads. All variables are expressed in basis points. Define l and m , respectively, as the maturities of the nearest-maturity LIBOR and Treasury for a given trade. The maturity-matched TED spread for the trade is then defined as $LIBOR(l) - Treasury(l)$ and the maturity-matched Treasury yield is defined as $Treasury(m)$; l does not equal m for longer-tenor trades (e.g., 30-year Treasury swap) because the maximum maturity LIBOR rate we observe is one year. t -Statistics are reported in parentheses under point estimates and are based on standard errors clustered by strategy-month. * and ** indicate statistical significance at the 10% and 5% level, respectively.

	Dep Variable: Δ Implied RF								
	Unsecured	Secured	CIP	Box	Equity S-F	TSwap	TFut	Tips-T	CDS-Bond
Δ Treasury	0.88** (9.34)	0.93** (51.66)	0.85** (5.66)	0.99** (16.68)	0.78** (2.85)	0.99** (60.79)	0.80** (8.74)	0.95** (35.97)	0.71** (9.81)
Δ TED	0.49** (4.58)	0.07 (1.33)	0.35** (2.24)	0.53** (5.26)	0.78** (3.81)	0.04 (0.94)	0.15 (1.35)	0.08 (0.77)	-0.24 (-1.12)
R^2	0.23	0.66	0.22	0.57	0.11	0.95	0.11	0.88	0.53
N	1,694	2,136	968	363	363	807	603	484	242

B. Shocks to Unsecured Funding and Arbitrage Activity

In this section, we show that variation in unsecured funding conditions induces comovement in unsecured arbitrage spreads but not secured spreads. We start with OLS evidence in Table III. We work with implied riskless rates from different arbitrages, as opposed to spreads that subtract out a benchmark riskless rate, to separate changes in secured and unsecured funding conditions. In the first two columns, we run the monthly panel regression

$$\Delta r_{i,j,t} = \alpha_{i,j} + \beta_1 \Delta y_{i,t} + \beta_2 \Delta TED_t + \varepsilon_{i,j,t}, \tag{10}$$

where $r_{i,j,t}$ is the implied riskless rate for individual trade i in broad strategy j in month t , $y_{i,t}$ is the yield on a Treasury with the same maturity as the horizon of the trade, which proxies for the true riskless rate, and TED_t is the maturity-matched Treasury-euro spread (i.e., LIBOR minus Treasury), which proxies for unsecured funding costs.²² Standard errors are clustered by strategy-month.

In the first column of Table III, the sample consists of unsecured trades (equity spot-futures, CIP, and box). These trades load on the Treasury yield with a coefficient close to 1, but also have a high loading on the TED spread, consistent with the idea that these trades require a significant amount of unsecured funding. Indeed, the coefficient on the TED spread of 0.49 is higher than the

²² The ICE Benchmark Association does not publish LIBOR rates beyond one year. Thus, when the tenor of the trade exceeds one year, we construct the TED spread using the one-year LIBOR and Treasury yields.

margin requirements listed in Table II, possibly because these trades require more unsecured funding on the margin than on average.²³

The second column of Table III shows a stark contrast for secured trades. These trades also load on the Treasury yield with a coefficient close to 1, but their loading on the TED spread is much lower (0.07 vs. 0.49) and is not statistically distinguishable from zero.²⁴ The remaining columns of Table III run the regression strategy-by-strategy. The coefficient on the TED spread is higher for all unsecured strategies than it is for any of the secured strategies. Moreover, we cannot reject the null that the TED spread loading is zero for each of the secured strategies, but we can for each of the unsecured strategies.

Previous research notes that arbitrage spreads are sensitive to the TED spread (e.g., Gârleanu and Pedersen (2011)), particularly during stress periods like the GFC. Our focus here is to highlight differences in the sensitivity of arbitrage strategies to the TED spread. We interpret these differences as showing that frictions in funding markets drive cross-sectional differences in arbitrage spreads.

While the results in Table III are consistent with funding segmentation, they could also reflect balance-sheet segmentation. For instance, suppose broker dealers specialize in unsecured trades. Then a deterioration in their balance-sheet health could lead to a simultaneous rise in the TED spread and unsecured arbitrage spreads. Formally, in the notation of the model, we interpret the results in Table III as differential loadings on a funding factor $f_{1,t}$, captured by the TED spread. However, they could also reflect variation in the marginal balance-sheet costs $c_{1t}V_{1t}$ of arbitrageurs who specialize in unsecured trades.

To isolate the role of funding segmentation, we follow Anderson, Du, and Schlusche (2019) and study the 2016 MMF reform. The reform modified SEC Rule 2a-7, which governs MMFs. It required Institutional Prime (IPrime) MMFs to switch from reporting stable to floating net asset values (NAVs), while allowing government MMFs to continue reporting stable NAVs. Thus, following the reform, many prime MMFs converted to government MMFs to accommodate client preferences for stable NAVs. Prior to the reform, prime MMFs were a significant source of unsecured funding for banks, so the reform plausibly represents a funding shock that is distinct from bank balance-sheet shocks.²⁵ Indeed, as shown in Figure 5(A), unsecured MMF lending to banks fell approximately \$550 billion as a result of the reform. Anderson,

²³ In Internet Appendix Section H, we provide suggestive evidence in favor of this interpretation for equity spot-futures arbitrage. We show that the value of equity securities held by dealers is nearly double the size of equity tri-party repo, which cuts against the idea that dealers fully finance their equity positions with equity repo.

²⁴ Note that correlations with the Treasury yield are very high for some secured trades because these trades involve Treasuries.

²⁵ Consistent with this interpretation, Figure 5 shows that the balance-sheet strength of dealers was not negatively affected by the reform. The measure of balance-sheet strength in the plot comes directly from He, Kelly, and Manela (2017) and is defined as the ratio of market capitalization to market capitalization plus book debt for the New York Federal Reserve's primary dealers' publicly traded holding companies. We discuss the distinction between balance-sheet and funding shocks in the context of the MMF reform in Internet Appendix Section F.

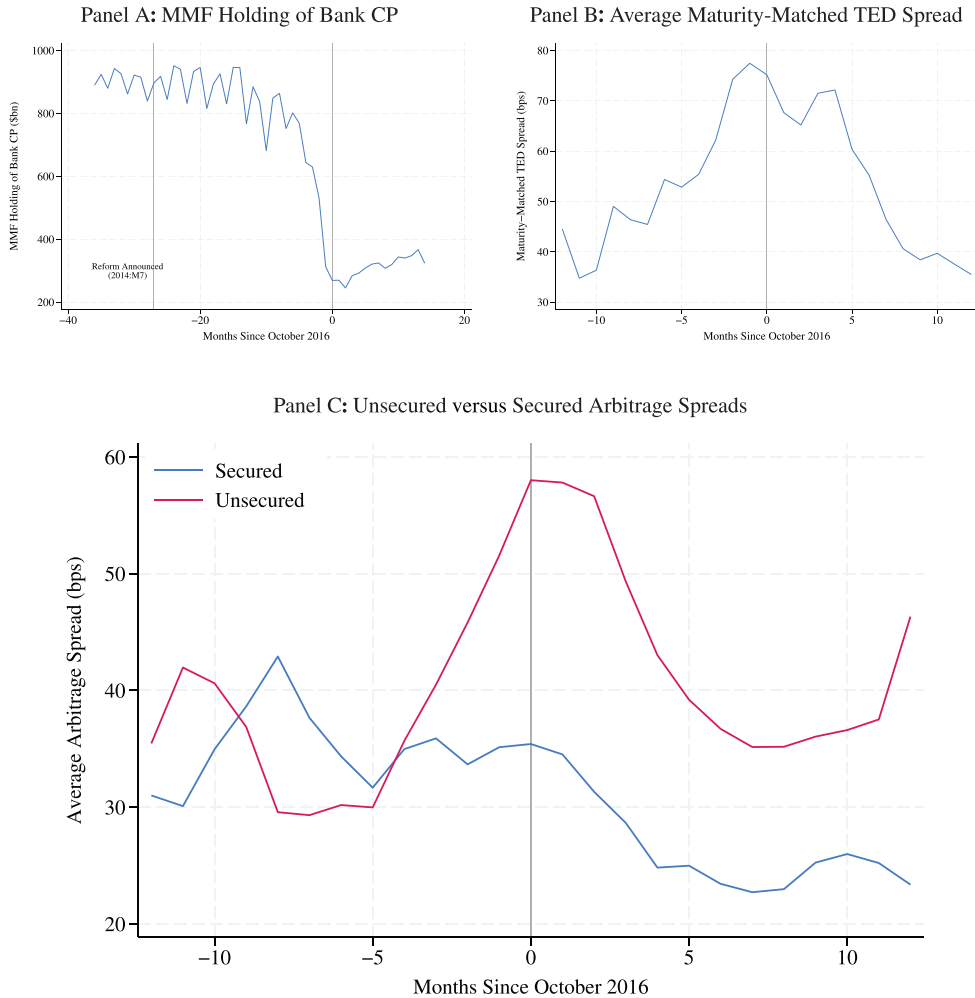


Figure 5. Event study of the 2016 money market reform. This figure summarizes money market fund (MMF) behavior, funding costs, and the absolute values of arbitrage spreads around the 2016 MMF reform. Compliance with the reform was required by October 2016 and thus we define the reform event as occurring in October 2016. Panel A shows the time series of bank commercial paper held by MMFs. Panel B shows the average maturity-matched TED spread (LIBOR - Treasury) for the arbitrages in our sample. Denote l as the maturity of the nearest-maturity LIBOR for a given trade. The maturity-matched TED spread for the trade is then defined as $LIBOR(l) - Treasury(l)$. For trades with tenors longer than one year, we set $l = 1$ year based on the availability of LIBOR rates. Panel C shows the average arbitrage spread of trades that rely heavily on unsecured funding (CIP, box, and equity spot-futures) and those that rely more on secured funding. (Color figure can be viewed at wileyonlinelibrary.com)

Du, and Schlusche (2019) study how global banks respond to this shock, arguing that they withdraw from CIP and central bank reserve arbitrage. In contrast, we use the shock to trace out funding segmentation in the cross section of arbitrage.

Figure 5(B) shows that the MMF reform shock generated a significant rise in the TED spread. As the reform was anticipated, spreads start rising before the reform is implemented. For example, five months before the reform, MMFs were more willing to lend to banks unsecured for four months than six months. Figure 5(C) shows that around the time of the reform, spreads on unsecured arbitrages rise relative to secured arbitrages. Thus, unsecured funding shocks induce comovement in arbitrage spreads for unsecured trades but result in low correlations between secured and unsecured trades.

Table IV provides formal regression evidence corresponding to these figures.²⁶ We first estimate the OLS regression

$$s_{i,t} = \alpha_i + \alpha_t + \beta 1[i \in Unsecured] \times 1[t \geq \text{October 2016}] + \varepsilon_{i,t}, \quad (11)$$

where $s_{i,t}$ is the absolute value of the arbitrage spread for trade i on date t , α_i is a trade fixed effect, and α_t is a time fixed effect. We estimate the regression using data through October 2017 to focus on the one-year impact of the reform on arbitrage spreads. Given the persistence of spreads at the daily level (Table I), we cluster standard errors by trade and time. In Internet Appendix Section G, we show that our inference is also robust to using Driscoll and Kraay (1998) standard errors, which are explicitly designed to handle serial correlation and heteroskedasticity in panel settings. Column (1) shows that unsecured spreads rose by an average of 12 bps in the year following the reform. In column (2), we estimate a dynamic version of equation (11) to more carefully study the response of spreads to the reform over time. Unsecured spreads initially rise 18 bps relative to other arbitrage spreads in October 2016 and remain elevated near that level for the subsequent three months, after which they only partially revert between February and October 2017. These findings indicate that the effect of the funding shock on arbitrage activity persisted for many months.

Furthermore, the pass-through of 0.59 implied by the 2016 MMF reform event study is similar to the OLS estimate of 0.49 in Table III, and we cannot reject the null hypothesis that they are equal.²⁷ This result suggests that most of the comovement between the TED spread and unsecured trades in our

²⁶ While Figure 5(C) and Table IV look similar to a differences-in-differences analysis, they are formally closer to a placebo test. In particular, the parallel-trends assumption should hold under the null of integrated funding. However, under our preferred interpretation—that the unsecured arbitrages are segmented from the secured arbitrages—there is no reason for the parallel-trends assumption to hold. That is, we do not think that the gap in spreads between unsecured and secured arbitrages would have remained fixed in the absence of the 2016 MMF reform. Instead, we simply interpret this evidence as indicating that only unsecured arbitrages are affected by a shock to unsecured funding.

²⁷ Around the reform, the TED spread and unsecured spreads increased by 31 bps and 18 bps, respectively. This implies that the pass-through of changes in the TED spread to unsecured spreads equals 0.59.

Table IV
Analysis of 2016 MMF Reform

This table shows estimates of the effect of the 2016 money market reform on the absolute values of arbitrage spreads. Column (1) presents estimates of the following daily regression: $s_{it} = \alpha_i + \alpha_t + \beta 1[i \in \textit{Unsecured}] \times 1[t \geq \textit{October2016}] + \varepsilon_{it}$, where s_{it} is the absolute value of the arbitrage spread for trade i on date t , $1[i \in \textit{Unsecured}]$ is a dummy variable that equals 1 if trade i relies heavily on unsecured funding (CIP, box, and equity spot-futures), and $1[t \geq \textit{October2016}]$ is a dummy variable that equals 1 on or after October 2016. Column (2) shows estimates of the regression $s_{it} = \alpha_i + \alpha_t + \sum_{j=-4}^3 \beta_j 1[i \in \textit{Unsecured}] \times 1[t = \textit{October2016} + j] + \beta_{j \geq 4} 1[i \in \textit{Unsecured}] \times 1[t \geq \textit{February2017}] + \varepsilon_{it}$. Arbitrage spreads are expressed in basis points. In column (2), we also report p -values for the null hypothesis that the coefficients prior to October 2016 (β_j for $j < 0$) are equal to zero, as well as the null hypothesis that the coefficients on or after October 2016 are equal to each other (β_j are equal for $j \geq 0$). All regressions include fixed effects for trade (α_i) and date (α_t). t -Statistics are reported in parentheses under point estimates and are based on standard errors clustered by trade and date. * and ** indicate statistical significance at the 10% and 5% level, respectively. The estimation sample ends one year after the reform in October 2017.

	Dep Variable: Arb. Spread (bps)	
	(1)	(2)
β	11.77** (2.47)	
$\beta_{j=-4}$		-4.01 (-0.59)
$\beta_{j=-3}$		0.80 (0.09)
$\beta_{j=-2}$		7.23 (0.82)
$\beta_{j=-1}$		11.72 (1.64)
$\beta_{j=0}$		18.03** (2.33)
$\beta_{j=1}$		18.29** (2.21)
$\beta_{j=2}$		20.70** (2.58)
$\beta_{j=3}$		16.29** (2.41)
$\beta_{j \geq 4}$		9.26** (2.18)
$p: \beta_j = 0, \forall j < 0$		0.00
$p: \beta_0 = \beta_1 = \beta_2 \dots$		0.00
Adjusted R^2	0.59	0.59
N	60,691	60,691

sample is driven by funding shocks, as opposed to bank balance-sheet shocks. Taken together, the analyses in Tables III and IV show that funding segmentation is one broad driver of low correlations among arbitrage spreads. Some trades—equity spot-futures, box spreads, and CIP—require more unsecured funding than others. These trades are therefore more exposed to broad conditions in unsecured funding markets, as measured by the TED spread. As a

result, unsecured trades tend to comove more with each other than they do with secured trades. In other words, funding segmentation impacts asset prices.

C. Further Funding Segmentation

We next provide evidence that funding markets are more segmented than the simple divide between secured and unsecured trades. In particular, we argue that additional funding segmentation helps explain why the equity spot-futures, box, and CIP trades, while more correlated than other trades, are still not highly correlated with each other. Building on the MMF literature (e.g., Chernenko and Sunderam (2014); Rime, Schrimpf, and Syrstad (2017); Li (2021); Hu, Pan, and Wang (2021)), we document that specialization in certain types of funding by MMFs is reflected in arbitrage spreads.

Our analysis starts from the observation made by Hu, Pan, and Wang (2021), who show that Fidelity MMFs were the largest provider of equity-repo financing during the 2010 to 2013 period. In Table V, we show that funding shocks to Fidelity move equity spot-futures arbitrage spreads over and above the effect of the TED spread. To do so, we augment equation (10) with flows into Fidelity's IPrime MMFs.²⁸ Columns (1) to (3) report OLS results. Column (1) shows that equity spot-futures arbitrage spreads fall when funds flow into Fidelity MMFs, consistent with the idea that a positive Fidelity funding supply shock reduces the cost of funding equity holdings and hence equity spot-futures spreads. Columns (2) and (3) show that flows to Fidelity have no impact on either other unsecured trades (box and CIP) or secured trades, suggesting that Fidelity funding supply shocks do not affect these trades. These results also suggest that flows into Fidelity are not proxying for aggregate unsecured funding conditions or aggregate intermediary balance-sheet health.

The OLS estimates in columns (1) to (3) are potentially biased because flows into Fidelity IPrime MMFs could be driven by supply or demand. On the one hand, when arbitrage spreads rise, arbitrageurs may demand funding from Fidelity IPrime MMFs to take advantage of the profit opportunity, leading to a positive correlation between spread changes and MMF flows. On the other hand, exogenous outflows from Fidelity IPrime MMFs will contract the supply of repo funding available for arbitrage and raise arbitrage spreads, resulting in a negative correlation between spread changes and flows. The opposing effects of supply and demand therefore work against finding any relationship between spread changes and Fidelity IPrime flows. Thus, the fact that there is still a negative relationship between changes in equity spot-futures arbitrage spreads and Fidelity IPrime MMF flows suggests that funding supply shocks dominate for this trade.²⁹

²⁸ The data for this analysis are based on mutual fund flows from the Center for Research in Security Prices (CRSP) and the SEC's Form N-MFP. According to Form N-MFP, virtually all of Fidelity's equity repo lending is done by its prime MMFs.

²⁹ Another concern is that the OLS results reflect balance-sheet segmentation. It could be the case that specific intermediaries are important for equity spot-futures arbitrage and that flows to

Table V

Arbitrage-Implied Riskless Rates and Funding Shocks to Fidelity

This table presents regression estimates of monthly changes in arbitrage-implied riskless rates on flows out of Fidelity IPrime money market funds (MMFs). The first three columns show OLS estimates, and the last three columns show IV estimates, where the instrument is net flows into all Fidelity MMFs interacted with the Fidelity IPrime share of assets, lagged by three months. We also include the change in the maturity-matched Treasury yield and the change in the maturity-matched TED spread. Define l and m , respectively, as the maturities of the nearest-maturity LIBOR and Treasury for a given trade. The maturity-matched TED spread for the trade is then defined as $LIBOR(l) - Treasury(l)$ and the maturity-matched Treasury yield is defined as $Treasury(m)$; l does not equal m for longer-tenor trades (e.g., 30-year Treasury swap) because the maximum maturity LIBOR rate we observe is one year. See Section III.C for details on instrument construction. Columns (1) and (4) show estimates using only equity spot-futures, columns (2) and (5) show estimates for other unsecured trades (CIP and box), and columns (3) and (6) show estimates for all secured trades. All implied riskless rates are in basis points and flows are in percentage points. t -Statistics are reported in parentheses under point estimates and are based on standard errors clustered by strategy-month. * and ** indicate statistical significance at the 10% and 5% level, respectively. The F -statistic from the first stage of the IV is reported at the bottom of the table.

	Dep Variable: Δ Implied RF					
	(1) Equity S-F	(2) CIP/Box	(3) Secured	(4) Equity S-F	(5) CIP/Box	(6) Secured
Δ Treasury	0.79** (2.94)	0.91** (9.65)	0.93** (51.57)	0.81** (3.01)	0.91** (9.45)	0.94** (50.11)
Δ TED	0.70** (3.56)	0.39** (3.24)	0.07 (1.35)	0.62** (3.16)	0.41** (3.20)	0.12* (1.95)
Fidelity IPrime Flows	-0.55** (-3.86)	-0.14* (-1.84)	0.01 (0.34)	-1.09** (-2.25)	0.04 (0.12)	0.27 (1.38)
Estimation	OLS	OLS	OLS	IV	IV	IV
First-Stage F				9	15	30
R^2	0.15	0.31	0.66	0.11	0.30	0.65
N	363	1,331	2,136	357	1,309	2,099

In columns (4) to (6), we try to address this potential endogeneity bias by instrumenting with “passive flows” into Fidelity IPrime funds, defined as

$$Z_t = F_t \times L_{t-3}^I,$$

where F_t is the flow into all Fidelity MMFs and L_{t-3}^I is the lagged share of Fidelity MMF assets that are in its IPrime funds. The validity of Z_t as an instrument for flows into Fidelity IPrime funds rests on two assumptions. The first is that the lagged share L_{t-3}^I is exogenous to arbitrageur demand for funding from Fidelity IPrime funds, which seems reasonable given the lag length of

Fidelity reflect the health of those intermediaries’ balance sheets. In this case, however, the most natural interpretation is that both balance-sheet and funding segmentation are at work. Flows to Fidelity reflect the health of particular intermediaries over and above the TED spread because Fidelity has funding relationships with those intermediaries.

one quarter. The second is that demand for arbitrage funding is exogenous to flows into all Fidelity MMFs (F_t). This assumption is likely to hold for equity-spot futures arbitrage because equity repo lending is a very small portion of Fidelity's overall MMF portfolio. During our sample, equity repo lending accounts for less 1% of all lending by Fidelity MMFs and never exceeds 3.1%. Thus, while Fidelity is large relative to the equity repo market (average lender share of 53%), it is unlikely that demand for equity repo funding drives flows into all Fidelity MMFs.

Consistent with demand inducing bias into the OLS coefficients, column (4) shows that the relationship between equity spot-futures arbitrage spreads and Fidelity IPrime flows is stronger when we instrument.³⁰ Importantly, the IV estimate is close to zero and is not statistically significant for other unsecured arbitrages (column (5)) and secured arbitrages (column (6)), further confirming that Fidelity IPrime funds play no special role for these trades. Overall, the evidence in Table V shows that funding is segmented even within the unsecured market—the cost of funding equity holdings moves independently of other funding costs.

Taken together, our results suggest that funding segmentation is an important driver of segmentation in asset prices. Unsecured trades are broadly segmented from secured trades because unsecured funding is segmented from secured funding, with the TED spread capturing these differences. Beyond the simple divide between secured and unsecured funding, there is additional segmentation, which appears to be driven by specialization among funding sources.

D. Factor Analysis

To complement our analysis of unsecured and secured arbitrages, we now conduct a simple PCA of the two groups. To start, we document that the first PC of unsecured arbitrages explains 51% of the daily variation in the level of spreads. Reinforcing the results in Section III.B, Figure 6(A) shows that it is natural to interpret this PC as capturing common unsecured funding costs because the first PC closely tracks the TED spread, also shown in the figure. The correlation between the two series is 69% at the monthly level. The existence of a meaningful factor structure among unsecured arbitrages explains why they are more correlated with each other than secured arbitrages, yet it is also important to note that half of the total variation in unsecured spreads is not captured by common funding costs. As we argue in Sections III.C and IV, this remaining variation is likely driven by strategy-specific funding costs or balance-sheet shocks to specialized intermediaries.

A similar analysis of secured arbitrages indicates that the first PC explains 40% of the daily variation in spread levels. Figure 6(B) plots the monthly version of this PC along with the TED spread for reference. Unlike unsecured arbitrages, the first PC of secured arbitrages is largely unrelated to the TED

³⁰ We report the first stage in [Internet Appendix Section I](#).

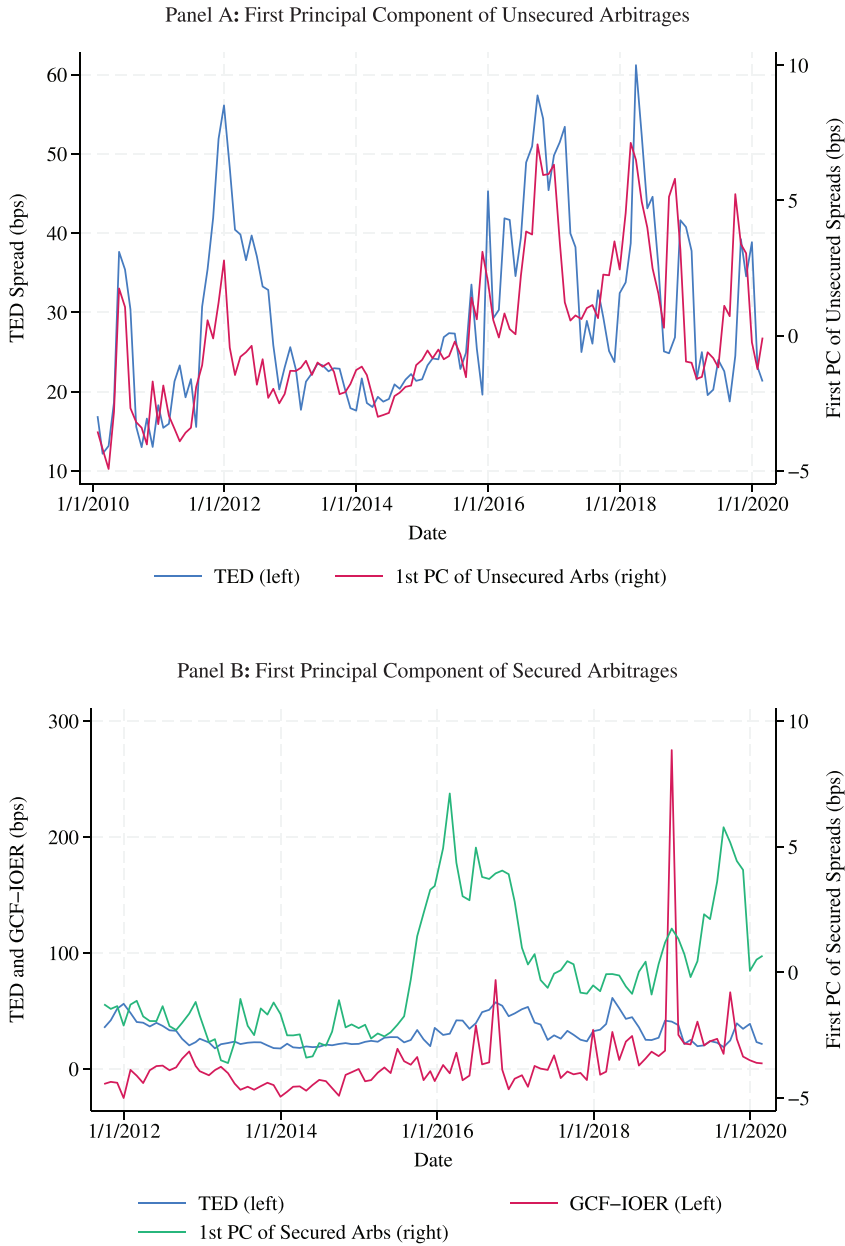


Figure 6. Funding costs and arbitrage correlations. Panel A shows the first principal component (PC) of unsecured arbitrage spreads and the three-month TED spread. Panel B shows the first PC of secured arbitrages, the one-day, one-week, and one-month tri-party general collateral finance repo rate (GCF) relative to the interest rate on excess reserves (IOER), and the three-month TED spread. All data are monthly. Panel B begins in September 2011 due to the availability of the 10- and 20-year Treasury-swap rate. (Color figure can be viewed at wileyonlinelibrary.com)

spread, as the two are only 36% correlated at the monthly level. This finding supports our argument that funding segmentation drives down the correlation between unsecured and secured arbitrages (Section III.B). The next question is how to interpret the first PC of secured arbitrages. We explore three possibilities: (i) secured funding costs, (ii) balance-sheet costs of dealers, and (iii) balance-sheet costs of nondealer intermediaries.

A first possibility is that secured funding costs move independently of unsecured funding costs, but also have a strong factor structure. If this were the case, the first PC of secured arbitrage spreads would have a high correlation with common proxies for funding conditions in secured financing markets. We find little evidence consistent with this hypothesis. For example, Figure 6(B) also plots the spread between the GCF repo rate and the IOER paid by the Federal Reserve. This proxy has been used to measure repo market conditions in several recent papers, including Correa, Du, and Gordon (2020) and Copeland, Duffie, and Yang (2021). It is clear from the plot that the first PC of secured arbitrages and the GCF-IOER spread follow different time-series trends, as their monthly correlation is only 30%.

Another possibility is that the first PC of secured arbitrage spreads reflects variation in the balance-sheet costs of banks, which could matter, for instance, because dealers intermediate secured funding markets, namely, repo. However, we think it is unlikely that the first PC of secured arbitrages reflects bank balance-sheet costs because the magnitudes do not line up. For instance, He, Nagel, and Song (2022) find that the overnight spread between the GCF and TPR spikes by roughly 20 bps at quarter-end, implying one-month spikes of 0.7 bps. By comparison, the time-series volatility of secured arbitrages is roughly 22 bps.

Thus, our preferred interpretation is that the first PC of secured arbitrages reflects common variation in the balance-sheet costs of (nondealer) arbitrageurs. Though such costs are inherently hard to measure without data on the exact institutions that are active in each trade, our analysis in Section IV.D supports this interpretation. In particular, we show that when hedge funds that specialize in fixed-income arbitrage experience poor returns in month t , spreads on secured arbitrages tend to increase in month $t + 1$, whereas unsecured arbitrages are not affected.

IV. Segmented Balance Sheets

We next provide evidence of a second driver of segmentation in asset prices: balance-sheet segmentation across intermediaries. As discussed in Section I, if different intermediaries specialize in different trades, then the tightness of their individual balance-sheet constraints will affect some arbitrage spreads but not others.

We provide three complementary analyses. First, we provide suggestive evidence from CFTC quantity data that different intermediaries are more central for different trades. We then examine two event studies: JPMorgan's London

Whale episode in 2012 and Deutsche Bank's exit from the CDS market in 2014. Finally, we show that the tightness of fixed-income hedge fund balance-sheet constraints is important for certain secured trades.

A. Suggestive Evidence from Quantities

Table VI uses CFTC data to provide suggestive evidence that different intermediaries play varying roles in certain arbitrage trades. The CFTC summarizes positions in different futures of different types of intermediaries: dealers, hedge funds (labeled by the CFTC as "leveraged funds"), and asset managers. For each intermediary type and contract, the CFTC reports total gross positions long and short of the intermediary type in the contract, as well as total positions in the contract netted by intermediary type. The data are silent on the specific intermediaries that are active in a particular trade, and therefore do not perfectly reveal the marginal price setter for each contract. The data do, however, give us a sense of which intermediaries are most active in which contract.

We compute three different measures of activity. First, we look at an intermediary type's gross share of activity in a contract—the sum of the intermediary type's long, short, and spread positions in that contract, divided by the total long, short, and spread positions in the contract. Second, we net within each intermediary type, taking the difference between gross long and gross short positions for the intermediary type. We then report the intermediary type's net position as a fraction of the total net positions across intermediaries. Finally, we report the fraction of days the intermediary type's net position is in the direction that would earn the arbitrage spread. A high fraction of days earning the spread would suggest that the intermediary type is an important arbitrageur for the contract, accommodating demand from other sectors.

All three measures tell the same story. Dealers are the biggest players in equity futures, while hedge funds and asset managers play a more important role in Treasury futures. For instance, dealers are in a net position that earns the arbitrage spread in equity futures on 87% of days, while hedge funds are in a net position to earn the spread on 45% of days and asset managers are in a net position to earn the spread on only 8% of days. Moreover, dealers have the largest share of equity futures in terms of gross and net positions. In contrast, hedge funds appear to be the most active in Treasury futures, as their net position earns the arbitrage spread on 58% of days. Dealers are in a net position to earn the arbitrage spread on 50% of days, although their shares of gross and net outstanding are relatively small compared to hedge funds and asset managers.

While certainly not definitive, these numbers suggest that dealer balance-sheet constraints are likely to be particularly important for equity futures, while hedge fund balance sheets are more important for Treasury trades. The notion that hedge funds are particularly active in Treasury spot-futures

Table VI
Trading Behavior in U.S. Futures Markets

This table summarizes the weekly positions of dealers, hedge funds, and asset managers using weekly reports on the Commitments of Traders provided by the Commodity Futures Trading Commission (CFTC). We use hedge funds (HF's) to designate traders classified by the CFTC as "leveraged funds." Gross positions by type are computed as the sum of long, short, and spread positions. Gross share is the percent of total gross positions outstanding across all reporting agents. The columns listed under Position Size (% of Net) are computed as follows: (i) compute the net position of each type i in week t as $Net_{it} = Long_{it} - Short_{it}$; (ii) compute the total net outstanding of the market Net_t by summing $|Net_{it}|$ across all reporting agents; and (iii) Position Size (% of Net) is then $|Net_{it}|/Net_t$. We include the CFTC's "Other Reporting" agents in our calculation of gross and net outstanding, but do not report their share in the table. This means that shares in the table will not sum to 100. Gross Share and Position Size are weekly averages for each contract. The columns under Earns Arbitrage show the percent of days on which the net position of the type would earn the observed arbitrage spread.

	Gross Share (%)			Position Size (% of Net)			Earns Arbitrage (% of days)		
	Dealers	HFs	Asset Mgrs	Dealers	HFs	Asset Mgrs	Dealers	HFs	Asset Mgrs
2-Year Treasury Notes	11	37	38	14	35	32	46	63	34
5-Year Treasury Notes	12	30	48	15	35	41	62	66	26
10-Year Treasury Notes	12	30	48	13	34	35	58	74	32
20-Year Treasury Bonds	12	25	56	28	15	47	37	45	62
30-Year Treasury Bonds	16	29	52	18	33	47	46	46	23
S&P 500 Index	21	30	41	29	19	46	88	98	1
Nasdaq Index	35	31	29	43	20	31	80	29	14
Dow Jones Industrial Average	52	32	11	48	31	16	94	8	9
Average Treasury	13	30	48	18	31	41	50	58	35
Average Equity	36	31	27	40	23	31	87	45	8

arbitrage is also consistent with Barth and Kahn (2021). We next turn to event studies for more definitive evidence.

B. Event Study: The London Whale

In this section, we first provide suggestive evidence that JPMorgan is a particularly important intermediary for equity spot-futures arbitrage. We then examine the impact of balance-sheet shocks to JPMorgan on equity spot-futures arbitrage spreads. According to Coalition Greenwich, a subsidiary of S&P that provides benchmarks for the financial services industry, JPMorgan has had the largest share of the market for equity derivatives since 2015.³¹ This accords with data from bank regulatory filings that provide further suggestive evidence. In particular, we use the Y-9C regulatory filings to examine the trading book securities holdings of all U.S. bank holding companies. JPMorgan had by far the largest holdings of equity securities in its trading book over our sample, accounting for 37% of the total. JPMorgan's dominance was greater earlier in the sample. For instance, it held 56% of all equities in trading books in 2010. This evidence suggests that JPMorgan could play an outsized role in equity spot-futures arbitrage.

We now turn to the impact of an exogenous balance-sheet shock to JPMorgan—the so-called “London Whale” episode—on equity spot-futures arbitrage spreads. The London Whale episode was a result of activities by JPMorgan's Chief Investment Office (CIO) designed to hedge credit risk in the bank's loan portfolio. The Senate Permanent Subcommittee on Investigations issued a detailed report on the episode, from which we draw the following background information.³² At the beginning of 2012, JPMorgan wished to reduce the size of its hedges in the credit derivatives market. Rather than simply exiting its existing positions, the CIO instead sought to offset the credit protection it had bought by selling credit protection. In doing so, it became one of the biggest players in credit derivatives markets, with other traders nicknaming it the London Whale. In addition, it incurred significant basis risk, in terms of both the credit quality and maturity of the credit protection it had bought versus sold.

As shown in Figure 7(A), this risk-taking resulted in significant losses, which reached over \$6 billion by the end of 2012. For context, the firm's market capitalization at the time was about \$125 billion. Figure 7(A) shows that losses began to accelerate in March 2012, with monthly losses totaling \$550 million and representing 75% of the firm's year-to-date losses. The Senate report also indicates that several internal risk limits were breached for the first time during the month. Another important event occurred on June 13, 2012, when JPMorgan CEO Jamie Dimon testified before Congress and

³¹ The full report can be found at <https://www.bloomberg.com/news/articles/2020-03-26/jpmorgan-s-equity-derivatives-haul-said-to-soar-to-1-5-billion?sref=wacugGYd>.

³² The report is available at [https://www.hsgac.senate.gov/imo/media/doc/REPORT%20-%20JPMorgan%20Chase%20Whale%20Trades%20\(4-12-13\).pdf](https://www.hsgac.senate.gov/imo/media/doc/REPORT%20-%20JPMorgan%20Chase%20Whale%20Trades%20(4-12-13).pdf).

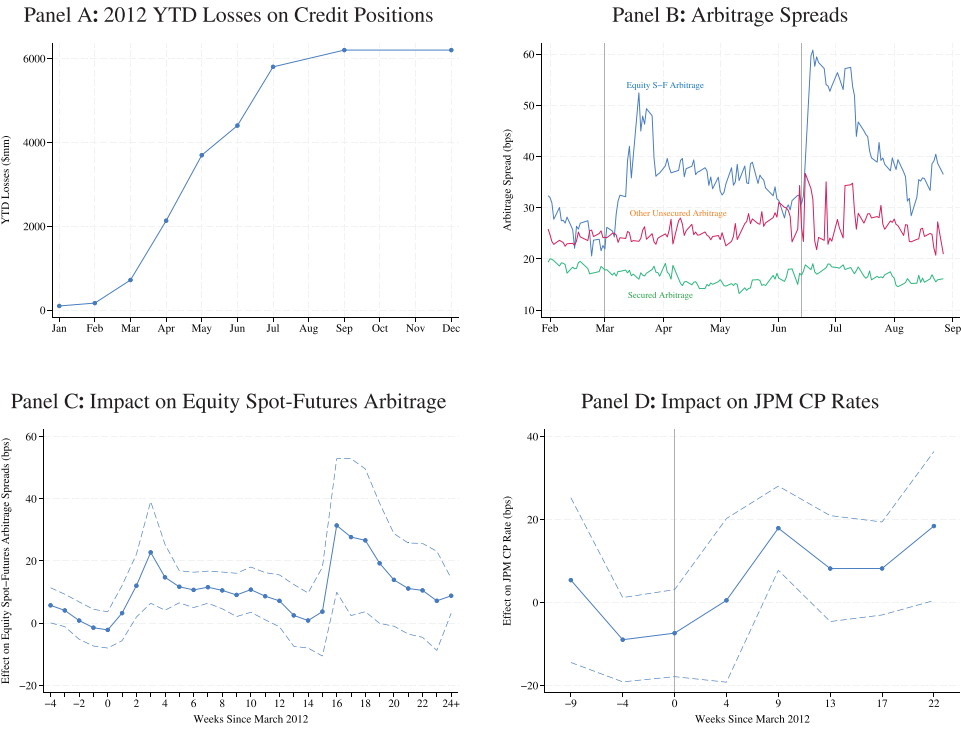


Figure 7. Event study of the 2012 JPMorgan London Whale. This figure summarizes JPMorgan's (JPM) losses, the absolute value of equity spot-futures arbitrage spreads, and JPM commercial paper (CP) borrowing rates around the 2012 JPM London Whale incident. Panel A shows the 2012 year-to-date losses on JPM's credit derivative portfolio, as reported by the U.S. Senate investigation into the incident. Panel B shows the daily average arbitrage spreads of equity spot-futures, other unsecured arbitrages (CIP and box), and secured arbitrages in 2012. The first vertical gray line in the plot is March 1, 2012, which is when losses began to accelerate. The second gray line is June 13, 2012, the first day that the CEO of JPM appeared before the U.S. Senate Committee on Banking, Housing, and Urban Affairs to testify about the Whale trades. Panel C shows the estimated impact on equity spot-futures arbitrage spreads relative to other unsecured arbitrages (CIP and box). The solid lines show the point estimates from a dynamic difference-in-difference model and the dotted lines show the associated 95% confidence intervals. Panel D shows the estimated impact on JPM's CP rate relative to the CP rates of other large global banks. See Section IV.B for more details. (Color figure can be viewed at wileyonlinelibrary.com)

announced that significant additional losses were to be expected at the firm's next conference call with shareholders. We therefore use March 1, 2012 and June 13, 2012 as the focal points of our event study.

Figure 7(B) shows that around these critical dates, equity spot-futures arbitrage spreads increased relative to other spreads. These results are consistent with the idea that JPMorgan is a particularly important intermediary for equity spot-futures arbitrage. Losses incurred in the London Whale episode tightened JPMorgan's balance-sheet constraints relative to other intermediaries, moving equity spot-futures spreads but not other arbitrage spreads.

Figure 7(C) provides formal regression evidence of the comparison between equity spot-futures arbitrage spreads and other unsecured-funding intensive trades. In a weekly panel of the absolute value of spreads on unsecured trades, we estimate

$$s_{i,t} = \alpha_i + \alpha_t + \sum_{j=-4}^{24} \beta_j 1[i \in \text{Equity Spot-Futures Arbitrage}] \times 1[t = j] + \varepsilon_{i,t}. \quad (12)$$

Figure 7(C) plots the coefficients β_j as well as 95% confidence intervals and shows that the patterns observed in Figure 7(B) are statistically significant. Equity spot-futures arbitrage spreads significantly increased compared to other unsecured arbitrage spreads following the event dates (March 1, 2012 and June 13, 2012) and remained elevated for several months.³³

Finally, to bolster the argument that these results are due to balance-sheet constraints and not funding costs, Figure 7(D) shows the evolution of rates on JPMorgan's commercial paper over the same period. There is little indication that short-term funding costs move substantially, which we take as evidence that the London Whale was primarily a balance-sheet shock. Taken together, this evidence suggests that JPMorgan is an important intermediary for equity spot-futures arbitrage, and shocks to its balance-sheet constraints disproportionately impact those trades. In other words, balance-sheet segmentation helps explain the low correlation of arbitrage spreads.

C. Event Study: Deutsche Bank's Exit from CDS

In our second event study, we examine Deutsche Bank's exit from the CDS market. As discussed in Wang et al. (2021), in late 2014 Deutsche Bank announced that it was exiting the single-name CDS market and sold a significant fraction of its CDS portfolio to Citigroup. Consistent with a substantial adjustment in Deutsche Bank's participation in the CDS market, the notional value of CDS contracts outstanding fell from 2 trillion euros in its 2013 annual report to 1.4 trillion in its 2014 annual report. The exact timing of Deutsche Bank's exit is unknown, but Bloomberg reported the sale to Citigroup in September 2014, and Deutsche Bank publicly announced the exit on November 17, 2014. Wang et al. (2021) study the effects of Deutsche Bank's exit on CDS market liquidity. In contrast, we are interested in its effect on CDS-bond arbitrage spreads, as compared to other arbitrage spreads.

Figure 8(A) depicts spreads around the exit event, which we center around the first week of October. Throughout late 2014, CDS-bond arbitrage spreads

³³ A potential confounding factor for the second event date (June 13, 2012) is that it is near the futures roll date in June. In Internet Appendix Section J, we show that the size of the increase in the calendar spread during this period is an order of magnitude larger than what is typically observed around futures roll dates, suggesting that the June 2012 increase is not mechanically related to contract rolling. Another potential concern with regression (12) is that the persistence of spreads $s_{i,t}$ distorts our inference of the β_j 's. We explore this issue in Section G of the Internet Appendix.

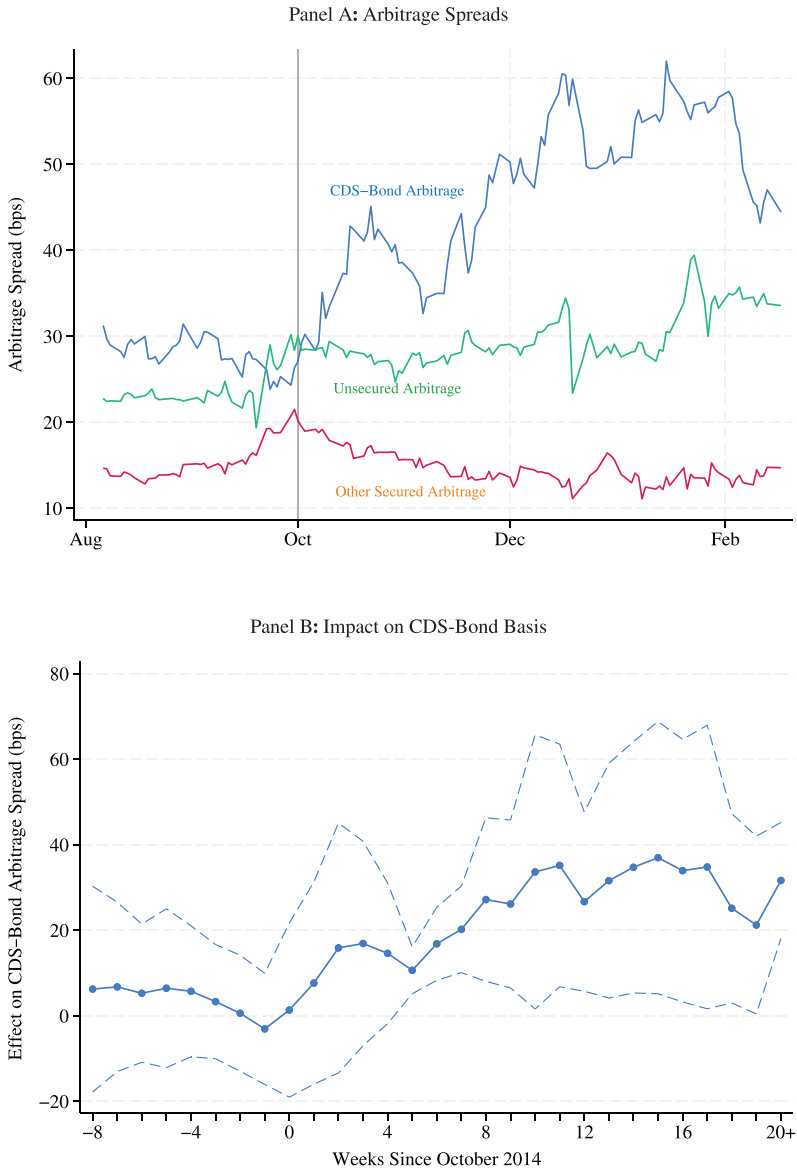


Figure 8. Event study of Deutsche Bank's 2014 exit from CDS trading. This figure summarizes the behavior of the absolute values of arbitrage spreads around the 2014 exit of Deutsche Bank (DB) from the CDS market. Panel A shows the daily average arbitrage spreads of CDS-bond arbitrage, other secured arbitrages (Treasury futures, Treasury swap, and TIPS-Treasury), and unsecured arbitrages in the last half of 2014 and the beginning of 2015. The gray vertical line in the plot is October 1, 2014. The exact timing of DB's exit is unknown, but there are reports that they sold a large portion of their CDS portfolio to Citibank in September 2014 and they publicly announced the exit on November 17, 2014. Panel B plots the point estimates from a dynamic difference-in-difference model and the associated 95% confidence intervals around the event. See Section IV.C for more details. (Color figure can be viewed at wileyonlinelibrary.com)

rise, but other arbitrage spreads do not. Figure 8(B) provides formal statistical evidence by running a regression analogous to (12) for the CDS-bond basis relative to other secured trades. The plot shows that the differential impact of Deutsche Bank's exit on CDS-bond arbitrage spreads relative to other secured spreads is significant at the 5% level. Furthermore, the relative widening of the CDS-bond arbitrage spread persisted for over five months. These results are consistent with the idea that Deutsche Bank was a particularly important intermediary for CDS-bond arbitrages. Its decision to exit the market is akin to a tightening of its balance-sheet constraints, which moved CDS-bond arbitrage spreads but not other spreads.

D. Hedge Fund Balance-Sheet Constraints

We next turn to the impact of hedge fund balance-sheet constraints, which we capture using monthly hedge fund returns. The idea is that following negative returns, hedge funds face tighter balance-sheet constraints. Arbitrage spreads should subsequently rise for trades in which hedge funds are important intermediaries. The following monthly regression tests this idea:

$$\Delta s_{i,t} = \alpha + \beta r_{t-1}^H + \varepsilon_{i,t}, \quad (13)$$

where $\Delta s_{i,t}$ is the change in the absolute value of the spread on trade i in month t and r_{t-1}^H is the monthly return of hedge funds that specialize in fixed income arbitrage at $t - 1$. Hedge fund returns are measured using Barclay's fixed-income arbitrage hedge fund index, which is based on the returns of funds aiming to profit from price anomalies between related fixed-income securities, including interest rate swap arbitrage, U.S. and non-U.S. government bond arbitrage, and forward yield curve arbitrage.³⁴ The monthly hedge fund return series is standardized to have mean zero and unit variance. Lagged returns are used in the regression to avoid any confounding issues with reverse causality, since returns should be negatively related to contemporaneous changes in spreads but not future changes.

The first column of Table VII shows that lagged fixed-income hedge fund returns do not predict future increases in unsecured arbitrage spreads. In contrast, the second column shows a negative forecasting relationship for future changes in secured arbitrage spreads. A one-standard-deviation return loss for fixed-income hedge funds forecasts a 0.7 bp increase in future secured arbitrage spreads. The remaining columns of Table VII reveal that the relationship is driven primarily by the link between fixed-income hedge fund returns and the Treasury-swap and CDS-bond arbitrages. Overall, these results support the idea that hedge fund balance-sheet constraints are more relevant for secured trades than unsecured trades. Internet Appendix Section L provides further evidence of hedge fund balance-sheet segmentation.

³⁴ See <https://portal.barclayhedge.com/cgi-bin/indices/displayHfIndex.cgi?indexCat=Barclay-Hedge-Fund-Indices&indexName=Fixed-Income-Arbitrage-Index> for more information.

Table VII
Fixed-Income Arbitrage Hedge Fund Returns and Arbitrage Spreads

This table shows regressions of monthly changes in the absolute values of arbitrage spreads on the lagged aggregate return of hedge funds that specialize on fixed-income arbitrage, as measured by Barclay's Aggregate Fixed Arbitrage Index. The aggregate return series is standardized to have mean zero and unit variance. The columns Unsecured and Secured pool strategies based on whether they rely on unsecured funding (CIP, equity spot-futures, and box). The remaining columns run the regression by strategy. *t*-Statistics are reported in parentheses under point estimates and are based on standard errors that are clustered by strategy-month. * and ** indicate statistical significance at the 10% and 5% level, respectively.

	Dep Variable: Δ Arbitrage Spread								
	Unsecured	Secured	CIP	Box	Equity S-F	TSwap	TFut	Tips-T	CDS-Bond
FI Arb HF Return _{<i>t</i>-1}	0.00 (0.01)	-0.66** (-3.04)	-0.10 (-0.16)	-0.34 (-0.56)	0.64 (0.65)	-0.40** (-2.45)	-0.44 (-0.89)	-0.49 (-1.09)	-2.33** (-2.72)
R^2	0.00	0.01	0.00	0.00	0.00	0.02	0.00	0.01	0.06
<i>N</i>	1,694	2,136	968	363	363	807	603	484	242

To summarize, the results from this section suggest that balance-sheet segmentation is important for explaining the low correlations of arbitrage spreads. Intermediaries appear to specialize in certain arbitrage strategies. Furthermore, when an intermediary that is important for one arbitrage suffers a balance-sheet shock, the spread for that arbitrage can move without significantly affecting other arbitrage spreads. The price effects of shocks to specialized arbitrageurs imply that intermediary balance sheets are segmented.

V. Discussion and Conclusion

A. Persistence of Segmentation

While our empirical results document that both funding and balance-sheet segmentation impact asset prices, they are less clear about how long this segmentation persists. Following market dislocations, capital will ultimately flow to profitable arbitrage opportunities; the question is how quickly (Duffie (2010); Duffie and Strulovici (2012)).

In [Internet Appendix Section C](#), we use the sign-restricted SVARs estimated in [Section II.C](#) provide some insight into this question. We cumulate weekly supply shocks estimated from the SVARs to examine whether supply shocks in different markets become more correlated at longer horizons. We find that the supply shocks do become more correlated over longer horizons. Although this suggests some amount of arbitrage capital flows across trades, the average 1% upper bound of the correlations of quarterly supply shocks is only 37%, pointing to a relatively persistent form of segmentation in arbitrage. This persistence is also apparent from our event studies of the 2016 MMF reform ([Section III.B](#)), the JPMorgan London Whale ([Section IV.B](#)), and Deutsche Bank's exit from the CDS market ([Section IV.C](#)). In all three events, a subset of arbitrage spreads remained elevated for several months after the initial funding or balance-sheet shock.

In [Internet Appendix Section K](#), we supplement our analysis by studying how the returns to different arbitrage strategies comove over varying holding periods. While correlations do increase over longer holding periods, they remain far from perfect ($\sim 15\%$) even for holding periods of one quarter. With that said, some parts of the market appear more integrated than others over longer horizons, such as the three-month GBP CIP and the 10-year Treasury swap trades.

B. Conclusion

In this paper, we show that riskless arbitrage is segmented. The average correlation between arbitrage spreads is low. We show that this low correlation is due to both funding and balance-sheet factors.

Overall, our results demonstrate the importance of both balance-sheet and funding segmentation in financial intermediation. In this respect, we build on research that documents how shocks to specialized risk-bearing capacity can disconnect risk premia across markets. Our focus on fundamentally riskless

arbitrage trades highlights the pervasiveness of these issues. The arbitrages that we study are relatively straightforward to execute and have expected returns that are essentially observable. These characteristics should mitigate the typical agency problems thought to underlie segmentation, slow-moving capital, and the limits of arbitrage, yet in practice arbitrage still appears fairly segmented. It seems natural to expect more segmentation in the intermediation of risky assets where agency problems are likely to be more severe. More broadly, our results suggest that exploring the boundaries of the firm for financial intermediaries—why certain trades are grouped together in a market segment—represents a promising direction for future research.

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Appendix S1: Internet Appendix. Replication Code.

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